What Does Reinforcement Learning Add to a Small Translator for African Languages?

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Abstract

Small language models translate many African languages poorly, and one common failure is that they do not finish their sentences: they stop midway or repeat themselves. We train a small English-to-African translation model with supervised fine-tuning, then with reinforcement learning that rewards outputs rated well by automatic quality scorers, and we ask what the second step adds. It improves every language we test, by about nine points of the standard chrF++ score on one benchmark. About half of that gain is finishing sentences and not repeating; a single decoding setting gives the fine-tuned model the same half without any extra training, while the rest of the gain remains. Spending the same computing time on more fine-tuning helps only the languages for which we supplied extra data, and by the rule we fixed beforehand the comparison is inconclusive. For scientific text, the extra fine-tuning step, not the reinforcement learning, carries the gain. On 43 languages covered by a published system, our model reaches about 92% of the quality of NLLB-3.3B, which is about 8.6 times larger. We trained one model per setting, checked seed noise on a subset only, and used automatic scores only; no person judged the outputs.

1 Introduction

Machine translation into African languages is hard for small models. Training text is scarce, and small models have little capacity to spare. Large systems such as NLLB-200 (NLLB Team et al., 2022) help, but they are big, and the NLLB models and their outputs carry non-commercial licence terms.

Small models also fail in a characteristic way. After supervised fine-tuning (SFT), our model often does not finish its sentences: it stops midway, repeats a phrase until it runs out of space, writes in the wrong language, or copies the English input. Reinforcement learning (RL) with automatic quality scores as rewards is a natural remedy, and recent work reports gains for translation (Shao et al., 2024; Feng et al., 2025; Finkelstein et al., 2026). Those gains, however, may come from something simpler. Repeating-and-not-stopping is partly a property of greedy decoding, which a decoding setting can change; and RL costs computing time that more SFT could use instead.

We therefore ask four questions about a 385-million-parameter English-to-African translator, trained first with SFT and then with RL:

- **RQ1.** How much does RL improve translation, and for which languages?
- **RQ2.** How much of the gain is finishing and not repeating, rather than better translation, and can a decoding change provide it?
- **RQ3.** Does the same computing time spent on more SFT do as well?
- **RQ4.** How reliable is the gain, and how does the model compare with a much larger system?

Findings at a glance.

- RL improves every language we test on three standard benchmarks (RQ1).
- About half of the gain is finishing sentences and not repeating. A decoding setting recovers that half for the SFT model without training, but RL still adds clear gains beyond it (RQ2).
- Decoding tricks such as beam search and minimum Bayes risk (MBR) decoding help the SFT model a lot and the RL model little (RQ2).
- Matching the compute with more SFT is inconclusive by our pre-registered rule: it matches RL on the 19 languages where it had extra data, and is far behind on the others (RQ3).
- Almost all of the gain arrives in the first RL stage. The model reaches about 92% of the chrF++ of NLLB-3.3B on shared languages, at about one ninth of its size (RQ4).

Contributions. (1) A controlled decomposition of an RL gain in small-model translation:

a decoding control, a matched-compute control, seed and ablation runs, and an independent audit of the reward's own repetition check. (2) A decontaminated SFT and RL pipeline for 125 languages, with an audit of the training text. (3) Negative results: RL aimed at weak languages, science fine-tuning on a quality-estimation metric, and whole-document translation. Model artefacts: model weights at `https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl` (see Ethical considerations). We do *not* claim that RL beats more fine-tuning in general, that our model beats NLLB, or any human-rated quality: all scoring is automatic.

2 Related Work

Translation for African languages. Community benchmarks and corpora made progress measurable: Masakhane (Nekoto et al., 2020), MAFAND-MT (Adelani et al., 2022), FLORES-200 and its community successor FLORES+ (Goyal et al., 2022; Open Language Data Initiative, 2025), NTREX-128 (Federmann et al., 2022), SALT (Akera et al., 2022) and SMOL (Caswell et al., 2025). NLLB-200 (NLLB Team et al., 2022) is the reference system. Recent language models for African languages are much larger than ours (Yu et al., 2026; Akera et al., 2025); TranslatePsy-AfriSLM (Gritta et al., 2026) is the nearest small competitor. For science text, AfriScience-MT (Abdulmumin et al., 2026) uses SSA-COMET (Li et al., 2025) as its main metric.

Reinforcement learning for translation. GRPO (Shao et al., 2024; Yu et al., 2025) powers recent translation systems with reasoning-style training (Feng et al., 2025; Wang et al., 2025; Cheng et al., 2025; Zheng et al., 2025), and TranslateGemma rewards with an ensemble of quality models (Finkelstein et al., 2026). Closest to us, other work trains NLLB models with round-trip or reference-free rewards (Attia and Aji, 2026; Garcia-Estrada et al., 2026) and studies reward hacks in quality-estimation rewards (Liu et al., 2026; Han et al., 2026). Whether RL generalises beyond what supervised training memorises has been asked for reasoning (Chu et al., 2025; Yue et al., 2025). For translation, we searched for a comparison at matched computing time and found none; the nearest compares reference-free RL with three epochs of supervised training (Garcia-Estrada et al., 2026).

Decoding and evaluation. Minimum Bayes risk and quality-aware decoding (Freitag et al., 2022; Fernandes et al., 2022) aim at similar gains without training, so we measure them as controls. AfriCOMET (Wang et al., 2024), chrF++ and sacreBLEU (Popović, 2017; Post, 2018) and the paired bootstrap (Koehn, 2004) are our metrics and tests. Learned rewards are easy to over-optimise (Gao et al., 2023; Coste et al., 2024), which is why we keep the metrics we report separate from the metrics we reward. We claim no priority for RL in translation, for small models, or for language coverage.

3 Approach

Figure 1 shows the pipeline. We describe the model and the two training steps, then the variants used as controls and extensions.

Model. We start from Gemma-3-270m-it (Gemma Team et al., 2025) and add layers to reach 385M parameters (39 layers) (cf. Wu et al., 2024; Kim et al., 2023). The task is English to one of 125 languages. A single extra token in the vocabulary names the target language, and the model answers with that token followed by the translation.

Step 1: supervised fine-tuning (the SFT baseline). We train on English-to-African sentence pairs and instruction data. Before training, we remove every pair that matches the text of any benchmark we later use, checked on both sides of the pair. Part of the data comes from NLLB-mined sentences and a collection of translated instructions; we report their share and licences in the Ethical considerations. Training uses the Muon optimiser (Liu et al., 2025) and takes about 45 hours on one A100 GPU (Appendix B).

Step 2: reinforcement learning (the RL model). We use GRPO (Shao et al., 2024). For each English text the model samples 16 translations. Each translation receives a reward, and the model is nudged towards translations that score above the average of their group and away from those below it; a penalty keeps the model close to its starting point. Three stages follow one another, and the first does most of the work (Section 5.4). The reward has three parts. (i) A translation that fails a hard check scores the minimum: wrong or missing language tag, no end of text, a copy of the input, repetition, the wrong language, or an implausible length. (ii) Otherwise a quality score without a reference translation rates the out-

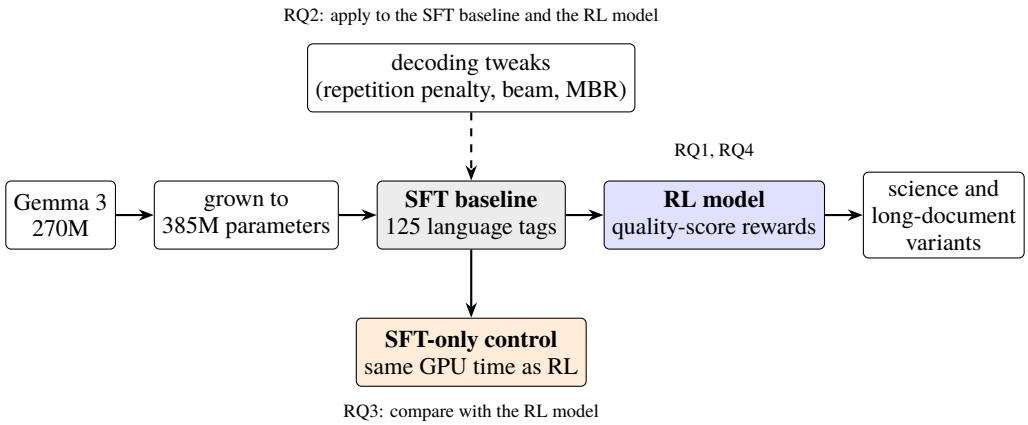


Figure 1: Overview. A 270M model is grown to 385M parameters and fine-tuned (the SFT baseline), then trained with reinforcement learning (the RL model). Two controls ask what else could give the same gain: decoding changes (RQ2) and more fine-tuning for the same GPU time (RQ3).

put. We combine two quality models (Wang et al., 2024; Li et al., 2025), taking their mean unless they disagree strongly, and for languages without a validated quality model we reward agreement with the group's other samples. (iii) Small penalties discourage, among other things, unusual length, missing numbers or names, and missing diacritics. Exact settings are in Appendix B.

Variants. The *SFT-only control* continues SFT from the baseline for the same GPU time as the RL stages, using half replay data and half extra English-to-X pairs for 19 languages (Gritta et al., 2026). The *science variant* adds a short fine-tuning step on science text from AfriScience-MT (Abdulmumin et al., 2026) and then a second RL run; the *long-document variant* adds RL on long inputs. Internal names used in the appendix tables are listed in Appendix A.

4 Experimental Setup

Benchmarks. We translate English into African languages on the standard FLORES+ devtest set (Open Language Data Initiative, 2025) (1,012 sentences; 54 languages), NTREX-128 (Federmann et al., 2022) (24 languages) and MAFAND-MT (Adelani et al., 2022) (15 languages). Smaller sets, SmolSent (Caswell et al., 2025), SALT (Akera et al., 2022), GlotStoryBook (Kargaran et al., 2023), IbomMT (Kalejaiye et al., 2025) and FLORES-MOZ (MOZNLP, 2026), appear in the appendix, as does AfriScience-MT (Abdulmumin et al., 2026) for science text. Unless stated, we report the average over languages, with each language counting equally.

Metrics. *chrF++* compares character and word n-grams of the output with a human reference; higher is better. *Held-out AfriCOMET* is a learned score that compares source, output and reference; we never use it as a reward. The *finished rate* is the share of outputs that end properly; the *loop rate* is the share that repeat themselves. Both are the checks our reward also uses, so they are not independent evidence of better translation. We computed chrF++ with sacreBLEU 2.6.0 (word order 2, otherwise defaults) after Unicode normalisation; the full signature string was not saved.

Avoiding self-deception. First, *independence*: the metrics inside the reward (the two quality models, agreement between samples, and the finished, repetition and language checks) are not evidence that RL helped. Our evidence is chrF++ and held-out AfriCOMET against human references (Appendix O). Second, *paired tests*: every difference is a paired bootstrap over test sentences with a 95% interval, which covers sentence variance but not training randomness. Third, *contamination*: besides filtering the training pairs, we scanned 26 million training samples against all benchmark text and found no benchmark sentence of three or more words (Ethical considerations list the residuals). Fourth, *seeds*: every model is one training run; we measured seed noise only for the first RL stage on 20 languages.

Decoding. All main results use greedy decoding with up to 768 new tokens (vLLM, bf16). The *repetition penalty* used as a control is a decoding setting that discourages repeated tokens; we chose its strength on a development set by a rule fixed in advance. Beam search and MBR decoding are described where they appear.

Table 1: RL model against the SFT baseline ("baseline"). chrF++ (top rows, per benchmark with the number of languages) and AfriCOMET: higher is better. Averages over languages, greedy decoding, 95% paired-bootstrap intervals in brackets. One training run per model.

	baseline	RL model	gain
FLORES+ (54 lang.)	22.85	31.66	+8.81 [+8.59, +9.03]
NTREX (24)	29.25	37.17	+7.91 [+7.69, +8.12]
MAFAND-MT (15)	35.77	40.78	+5.01 [+4.76, +5.23]
all three, all lang.	26.59	34.55	+7.96 [+7.82, +8.12]
AfriCOMET, all three	0.565	0.628	+0.063 [+0.062, +0.064]
FLORES+: ends properly	83%	99.5%	
FLORES+: repeats	21%	5%	

Table 2: chrF++ with and without decoding changes (higher is better; 95% intervals for the RL gain). Top: all three benchmarks. Bottom: a 20-language subset with 200 sentences each. Repetition penalty = a decoding setting that discourages repeated tokens; MBR = choose the sample most similar to the others.

decoding	SFT baseline	RL model	RL gain
greedy	26.6	34.6	+7.96 [+7.82, +8.12]
+ repetition penalty	30.5	34.8	+4.37 [+4.28, +4.46]
greedy (20 lang.)	32.3	39.9	+7.56 [+6.89, +8.19]
beam, 4 beams	36.0	39.1	+3.07 [+2.67, +3.46]
MBR, 16 samples	38.3	40.8	+2.52 [+2.26, +2.80]

5 Results

5.1 RQ1: How much does RL improve translation?

Answer: a lot, for every language we test. The RL model beats the SFT baseline on all three benchmarks (Table 1). Every language on FLORES+ and NTREX improves, and so does every MAFAND-MT language except Nigerian Pidgin, which does not change measurably. The gain is large where the baseline was weakest. On FLORES+ the baseline leaves about one output in six unfinished and repeats itself in about one in five; the RL model almost always finishes and rarely repeats (Table 1, lower rows).

Where RL does not help. For Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri, quality stays very low (chrF++ below about 14) and RL makes one problem worse: the model increasingly answers in English. For Tunisian Arabic, English output rises from about one fifth of outputs to almost half (Appendix F). These languages were not among the RL prompts.

5.2 RQ2: Is the gain just finishing and not repeating?

Answer: about half of it is, and a decoding setting recovers that half for the SFT baseline; RL still adds clearly more. We tested this in three ways. First, we cut the baseline's output at the first repetition, which leaves the translation untouched; this recovers 49% of the chrF++ gain. Second, we gave the baseline a repetition penalty during decoding, with no further training. It recovers the same 49%. With the penalty on both models, RL is still about 4.4 chrF++ ahead (Table 2, Figure 2). This control is imperfect: the best penalty strength was at the edge of the range we searched, and stronger values looked better on a development set but were never evaluated fully, so the baseline may be undertuned.

Third, we tried stronger decoding. Beam search and MBR help the baseline much more than they help the RL model. MBR gives the baseline a large boost and the RL model a small one; for beam search, the RL result depends on how the search decides to stop, so we draw no conclusion from it. RL with plain greedy decoding stays about 1.6 points ahead of the best decoded baseline, which needs about 17 times the computation per sentence. The beam-search numbers of an earlier study that used 256 output tokens are not comparable, and we do not use them (Appendix H).

A caveat on our own checks. The repetition flag we report is also part of the reward, so we remeasured with a detector that shares none of its code. It finds about 1.8 times as many repetitive outputs as the reward's flag does. Rescoring the extra outputs at their language's average changes the RL model's chrF++ by at most 0.2, so the conclusion stands (Appendix F).

5.3 RQ3: Does the same compute on more SFT do as well?

Answer: inconclusive by the rule we wrote down beforehand. We continued SFT for the same GPU time as the three RL stages, on half replay data and half extra English-to-X pairs for 19 languages. The extra pairs are NLLB-3.3B translations of web-text sentences (Gritta et al., 2026). We fixed the decision rule, the main comparison and the language groups before looking at any benchmark result for the control. The rule had four conditions; three were met. The fourth, that RL must also beat the control on the 19 languages that had extra data, was not met. So by our rule the result is inconclusive, and we did not change the rule afterwards.

In plain terms (Table 3, Figure 3): on the 19 languages with extra data, the control matches RL in chrF++ and finishes its sentences almost as reliably; it is even ahead on the learned metrics there, which were partly used to filter its data. On the

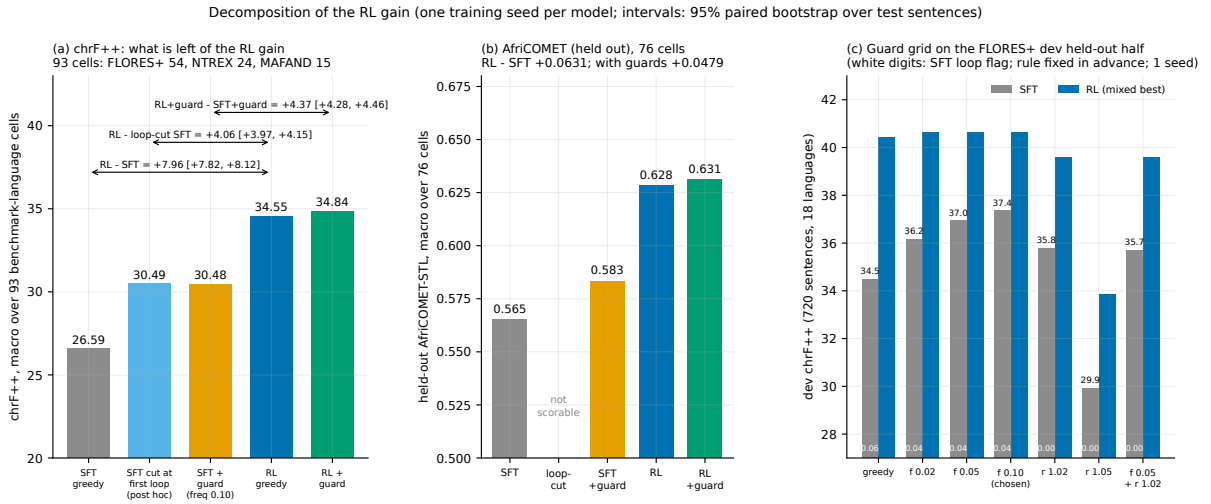


Figure 2: What is left of the RL gain once finishing is fixed. (a) chrF++, higher is better, averaged over all three benchmarks: the SFT baseline, the baseline with its repeated tails cut off, the baseline with a repetition penalty, the RL model, and the RL model with the same penalty. Arrows show paired differences with 95% intervals. (b) The same for held-out AfriCOMET. (c) How the penalty strength was chosen on a development set; the chosen value is at the edge of the grid.

Table 3: SFT-only control against the RL model, chrF++ (higher is better), averaged over language–benchmark pairs, 95% paired-bootstrap intervals. The control received extra data for 19 languages.

languages	control	RL model	RL − control
all	29.97	34.55	+4.58 [+4.45, +4.70]
19 with extra data	41.37	41.34	−0.03 [−0.13, +0.09]
all others	18.32	27.61	+9.29 [+9.09, +9.49]

Table 4: Comparison on the 43 languages that our evaluation shares with the published NLLB table. chrF++, higher is better. Differences with 95% intervals over languages.

system	chrF++	share	difference
SFT baseline	26.3	68%	−12.50 [−14.56, −10.40]
RL model	35.5	92%	−3.26 [−4.29, −2.20]
NLLB-3.3B	38.8	100%	

other languages, the control barely improves on the baseline, while RL improves them all. Taking only the overall number, RL looks better; taking only the 19 languages, the two look equal. Neither alone is the registered verdict.

Caveats. The control is one run with one learning rate and one stopping point; a different mix or a longer run could change the picture. It is also not data-matched: its extra pairs come from a larger system (NLLB-3.3B). We therefore claim neither that RL is a better use of compute than SFT nor the opposite.

5.4 RQ4: How reliable is the gain, and how does the model compare with a larger system?

Answer: the gain is robust to random seeds, nearly all of it comes from the first RL stage, and the model reaches about 92% of NLLB-3.3B's chrF++ on shared languages. The first RL stage produces 94% of the gain reached after the second stage, and the two later stages add little. Two extra random seeds of the first stage, run on 20 languages, differ by only 0.06 chrF++, far less than the gain itself. The last RL step is about the size of that seed noise, so we do not call it a gain (Figure 4).

We also tested four features that we added during the first stage. Together they are worth about 0.3 chrF++ at 100 steps, mostly because of one of them, a term that rewards agreement with human reference translations for seven languages. A beam-search guide, which costs about a quarter more time per step, is not shown to be worth it, and none of the features causes the drift from the starting model.

NLLB-3.3B is about 8.6 times larger than our model. RL closes about three quarters of the gap that the baseline leaves (Table 4, Figure 5), and the remaining gap is not a matter of unfinished sentences. The RL model matches or beats NLLB-3.3B on seven of the shared languages, all among those where NLLB scores lowest. Several caveats apply: NLLB's numbers are published ones on a slightly different test set and use beam search, our evaluation is greedy; the hardest languages have no published NLLB number; and for a few of the languages the reward itself used NLLB outputs as a target. We therefore do not claim to beat NLLB. Compared with our earlier 270M model run in the same harness, the 385M SFT baseline is behind, the RL model is level on two benchmarks and behind on one, and it is behind at document level (Ap-

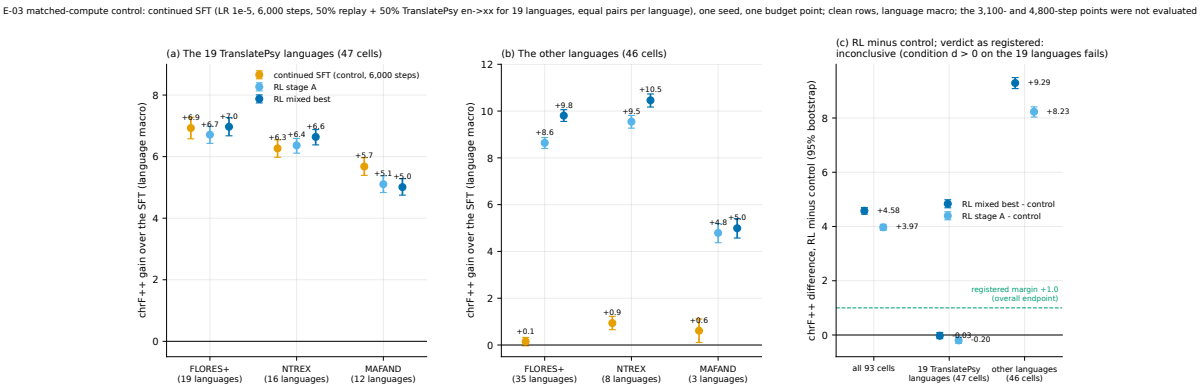


Figure 3: Gain over the SFT baseline in chrF++ (higher is better) for the SFT-only control, the first RL stage and the final RL model, on the 19 languages with extra control data (a) and on the others (b); (c) RL minus control, with the $+1.0$ margin of the registered rule. Error bars: 95% intervals. One run each; only the final control checkpoint was evaluated.

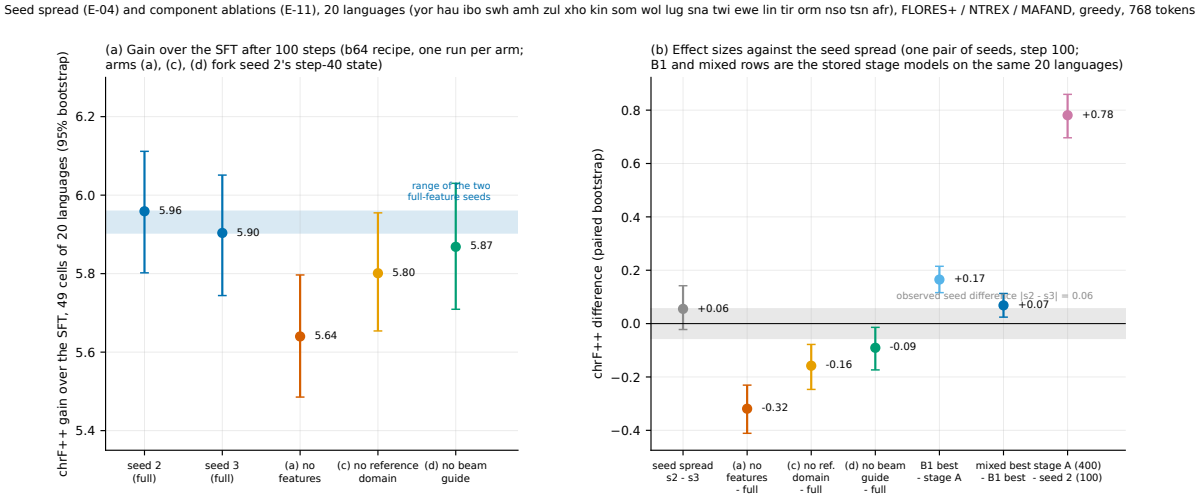


Figure 4: Seed noise and what four extra training features are worth (20 languages, chrF++ gain over the SFT baseline after 100 RL steps). (a) Two seeds of the full recipe (shaded band) and three runs that each remove a feature. (b) The same effects against the seed difference (shaded): removing all four features costs about 0.3 points, removing the beam-search guide is not separable from seed noise. Error bars: 95% intervals; one run per arm.

pendix C).

5.5 Further results: science text and long documents

Answer: for science text, the extra fine-tuning step carries the gain, and whole-document translation still does not work. A short fine-tuning step on science text raises chrF on AfriScience-MT but not the neural score the benchmark uses. An RL run started from that model adds a little more. To ask whether the science step is needed, we ran the same RL without it (Table 5): it ends about 1.4 chrF below the model that had the science step (one run each), and it gains little over its own start. The neural quality score rates the two models as equal, although chrF and held-out AfriCOMET rate the no-science-step model lower; because that score belongs to the family of scores the RL optimises, we do not treat it as evidence of quality. Translating whole documents in one pass fails for all our models. RL on long inputs raises the share of finished documents from 70% to 82%, but about three quarters of the documents still contain loops, so we translate sentence by sentence (Appendix K).

Table 5: Science text (AfriScience-MT test sentences, six languages, one run each). Higher is better. The last row is 1.44 chrF $[1.25, 1.61]$ below the third row.

model	chrF	AfriCOMET
RL model	45.4	0.683
+ science fine-tuning	46.6	0.688
+ science fine-tuning, then RL	47.1	0.693
RL without science fine-tuning	45.7	0.685

6 Discussion and Conclusion

What the evidence says. RL improves a small translator on every benchmark we use, but the gain has two parts that deserve separate names. About

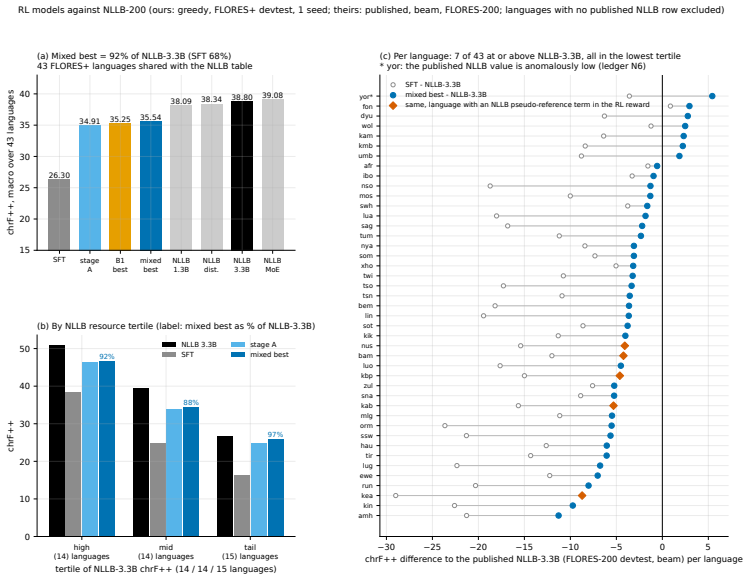


Figure 5: The SFT baseline and the RL model against NLLB-200 on the 43 shared languages (chrF++, higher is better). (a) Averages. (b) By how well NLLB-3.3B does on each language (three groups). (c) Difference per language; orange marks languages where NLLB outputs were part of the reward. Intervals are over languages only.

half is finishing sentences and not repeating, which a decoding setting also gives the fine-tuned model; the other half is better translation that the setting does not provide. Almost all of it arrives in the first RL stage. More fine-tuning for the same time is a serious alternative where it has extra in-language data, and a much weaker one where it does not; our registered comparison is therefore inconclusive, and what RL adds in our data is breadth, since it improved every language from the model's own samples and a quality score. For science text, the fine-tuning step, not RL, carried the gain.

What we do not claim. We do not claim that RL is a better use of compute than SFT in general, that the improvement beyond finishing is quality a human would confirm, that our model beats NLLB or our earlier 270M model, or that any reward ingredient other than the four tested features helped. The model is not usable for Arabic dialects, Tamazight, Dinka, Tamasheq or Kanuri.

What would settle the open points. Evaluating the stronger repetition penalties and the two unevaluated control checkpoints, more seeds for the later stages, and human judgement of outputs. Together with the licence work in the Ethical considerations, these are the next steps.

Limitations

Automatic scoring only. The references of FLORES+, NTREX-128 and MAFAND-MT are human translations, but all scoring is automatic: no human judgement of system outputs was collected. The neural scores belong to one family, and AfriCOMET has a length and entity bias. **One training run per model.** We measured seed noise only for the first RL stage, on 20 languages and 100 steps, with one pair of extra seeds. The last RL step and per-language differences below about half a point cannot be separated from seed noise. A second seed of the SFT baseline is impractical (about 45 GPU-hours). **The matched-compute control is one run:** one seed, one learning rate, one data mix with NLLB-3.3B targets and one evaluated stopping point. The repetition-penalty control chose its best value at the edge of the range searched. **Rewarded metrics are not evidence.** The finished, repetition and language checks, and the quality scores the RL optimises, are part of the reward. The independent repetition detector shares the idea of the reward's check and cannot see non-repetitive failures. **Ablations are limited:** three arms, 100 steps, one run each, 20 languages; later RL stages and most reward ingredients were not ablated, and mid-run changes (Ethical considerations) mean the stages are not one clean design. **Decoding and engine.** Headline numbers use one inference engine in bf16. The two engines differ in bf16 but agree in fp32; we did not run the RL model in fp32. Beam search and MBR cover 20 languages and one beam width. **Coverage and comparisons.** Fifty-one of the 125 languages have no reference test and are unscored. The model fails on Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri, where RL increases English output. Comparisons with the earlier 270M model and with NLLB mix differences in data, steps, test sets and decoding. Document results rest on 28 papers per language, and whole-document translation remains unusable. **Not run.** Human judgement of outputs; the NLLB-data ablation; scans of packed training shards for contamination; a 54-language decoding study; a science fine-tuning step at full strength followed by RL; a second seed of the science run; long-document RL from the no-science-step model.

Ethical considerations

Data and licences. Part of the fine-tuning data is derived from NLLB. Sentences mined with NLLB make up about 14% of the tokens of the main fine-tuning stage; adding NLLB-Seed and the Aya translations (NLLB-3.3B outputs) gives about 19%, and adding machine-translated Tumbuka instructions (system not stated, possibly NLLB) about 23%. NLLB models and outputs carry non-commercial (CC-BY-NC) terms, which limits commercial use of the NLLB-derived data and signals; a licence statement must accompany any release of the weights. Several sources have unspecified or unconfirmed licences (the `*_mt560` OPUS sets, which are mostly Bible and Jehovah's Witnesses text and 69% of the main pair data; `multi-open`; AfroLingu-MT; an Efik New Testament sheet) and need a legal decision. IbomMT is CC-BY-NC-4.0 and is a disclosed exception. Religious text dominates many low-resource languages. Reward-side data: NLLB-3.3B outputs (CC-BY-NC) served as a reward target for five languages only, and RL prompts (FineWeb-Edu, Wikipedia, news including BBC News, which declares no licence and carries BBC copyright) are input text only. The Gemma terms of use apply to the derived weights, and pre-training exposure cannot be excluded. Reward and monitor models, as declared on the Hub (read 2026-11-05): the two SSA-COMET and the two AfriCOMET checkpoints are apache-2.0; GlotLID declares `license: other` with `license_name: apache-2.0-plus-notices`.

Release and licence. The final RL model (branch `main`) is at https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl, the SFT model at https://huggingface.co/AletheiaAING/gemma-0.4B-translate-sft, and the seed, ablation and science-variant weights at https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl-experiments; the control checkpoint is not released. At the last check the Hub repositories declare no licence field and have no model card; the Gemma terms of use apply, and the NLLB-derived and non-commercial data above limit commercial use **[OWNER: confirm release licence]**. The data repositories named in our reports were private at the last check **[OWNER: which parts of the data are released]**.

The control's extra data. For the SFT-only control we used TranslatePsy English-to-X pairs (CC-BY-NC-4.0; NLLB-3.3B translations of web sentences, filtered by AfriCOMET, SSA-COMET and MetricX) for that experiment only. No TranslatePsy row is in the SFT baseline, any RL stage, the science or long-document variants, or any released weight; the control checkpoint is neither released nor used in production. A good control result partly means "distilling NLLB". The NLLB-218 language-identification model (CC-BY-NC) was used for measurement only.

Decontamination and its residuals (the clean 385M pipeline). The audit found no benchmark sentence of three or more words in the audited training text. The remaining benchmark matches are one- or two-word strings, numerals and names (for example 0.25% of NTREX rows and 0.47% of MAFAND-MT rows). The FLORES+ development half (440 sentence ids) is in training on purpose, and the held-out half is protected. A lexical fuzzy scan of the pair and instruction files found no benchmark string of eight or more tokens as a contiguous run, but strings of four to seven tokens occur inside longer training text, and 263 benchmark strings of eight or more tokens overlap training text by character 5-grams; these are candidates, and we did not measure their effect. The first RL stage drew 6 flagged prompts of 28,120 from seven unfiltered prompt files, and 0.9% of AfriScience-MT training rows have their African-language sentence in the earlier pair data (the test papers are untouched). We did not run paraphrase, machine-translation round-trip or packed-shard scans. These statements are about the clean 385M pipeline. Our earlier 270M pipeline trained on text that held the English side of 95% of FLORES+ devtest; its numbers appear only on rows clean for both pipelines.

Analysis decisions and process. We chose checkpoints on a 720-sentence FLORES+ development subset, chose the languages for the weak-language run from devtest-derived tables, ran the decomposition and family analyses after seeing results, and changed the science headline metric after writing the interpolation rule. The authors overrode two failed training gates. The first RL stage was stopped after 40 steps and resumed with four added features; later stages were restarted once or twice, and the hacking alarm was loosened once. The science RL step cap was extended by the trainer, one long-document run lost most of its time to a disk-full crash loop, and the no-science-step run had one out-of-memory restart (replayed steps use the later records). LibreTexts rows were removed from the long-document pool before it started.

Intended use, risks and dual use. Outputs are machine translations of unverified quality, with documented failure modes: Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri are not usable, RL increases English output there, and documents should be translated sentence by sentence. The model should not replace NLLB-class systems or be used where mistranslation is costly (health, legal, safety). As with any translation model, it can produce text at scale in under-served languages, including spam or misleading content, and we release no content filter.

Author contributions

[Placeholder, to be filled by the lead author: one sentence per person describing what that person contributed (for example design, data, training, evaluation, writing). Authorship is based on contribution.]

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A Glossary of internal names

The main text uses descriptive names. The appendix tables use the internal names of our experiment log; this table maps them.

internal name	name in the main text	meaning
A0	earlier 270M model (base of the grown model)	the 270M model with six inserted layers, trained before growth
SFT-1, SFT-2, stage B	SFT baseline (training steps)	SFT-1 trains only the new layers and language tags; SFT-2 (stage B) trains all weights
SFT, SFT final	SFT baseline	the 385M model after supervised fine-tuning
stage A	first RL stage	400 steps on short English windows
B1 best, B1 final	second RL stage (best checkpoint at step 50; final at step 135)	RL on longer windows; best and final differ by less than the noise (E-20)
mixed, mixed best	third RL stage; **RL model**	short and long windows; its best checkpoint is the final general RL model used throughout
weak-group RL	RL aimed at eight weak languages	85 extra steps; no measurable gain on its targets
science SFT, α	science fine-tuning step; + *science fine-tuning*	fine-tuning on science text, then weight interpolation between the starting model ($\alpha = 0$) and the fine-tuned model ($\alpha = 1$); $\alpha = 0.5$ is used
science RL	science RL	RL started from the science fine-tuned model
E-14	science RL without the science fine-tuning	the same RL started from the RL model
long-document RL	long-document variant	RL on inputs of up to 2,048 source tokens
clean rows	rows used for comparisons	benchmark rows minus 214 rows flagged by the contamination audit
cells	language–benchmark pairs	one language on one benchmark; 93 pairs on FLORES+, NTREX and MAFAND-MT; 76 with a held-out AfriCOMET score
TP languages	the 19 languages with extra control data	TranslatePsy languages of the SFT-only control
reward flag, independent flag	the reward's repetition check; the independent repetition detector	see Appendix F
E-01	repetition-penalty control	decoding setting that discourages repeated tokens
E-02	beam search and MBR study	decoding at 768 tokens on 20 languages
E-03	SFT-only control	matched-compute continued SFT
E-04, E-11	seed pair and ablations	extra seeds and feature-removal runs of the first RL stage
E-05	neural scores for all stages	SSA-COMET-QE, AfriCOMET and chrF on AfriScience-MT
E-09, E-09b	engine and precision checks	vLLM against Hugging Face, bf16 against fp32
E-13	earlier 270M models in the same harness	re-run on rows clean for both
E-15, E-16, E-20	long-document, science and B1-final evaluations	evaluation runs
E-10, E-17	human evaluation; NLLB-data ablation	not run

B Hyper-parameters, recipe and reward details

This appendix collects the SFT and RL settings that Sections 3 and 3 summarise.

B.1 SFT recipe and learning-rate schedule

Recipe. Samples are packed into 4,096-token packs (one language per pack, block-diagonal attention) at 32 packs = 131,072 tokens per step; each pack is used once. Optimiser: Muon (momentum 0.95, 5 Newton-Schulz steps) on the 2-D decoder matrices and AdamW (betas 0.9/0.95) on embeddings and norms, weight decay 0.01, gradient clipping 1.0, 3% linear warm-up then cosine decay, seed 1234. The learning-rate schedule, read from the executed per-shard configs, is: SFT-1 (`g1a`, 1,500 steps) peak 5e-5, cosine to 50% of the peak (2.5e-5); shard 1 (`g1b_s1`) peak 5e-5, cosine to 80% (4e-5); shards 2-5 (`g1b_s2`..`g1b_s5`) peak 3e-5 (lowered from 5e-5 on 2026-11-01 because of forgetting in tiny languages), cosine to 80% (2.4e-5); shard 6 (`g1b_s6`) peak 3e-5, cosine to 10%, so the last step runs at 3.0e-6. Every shard

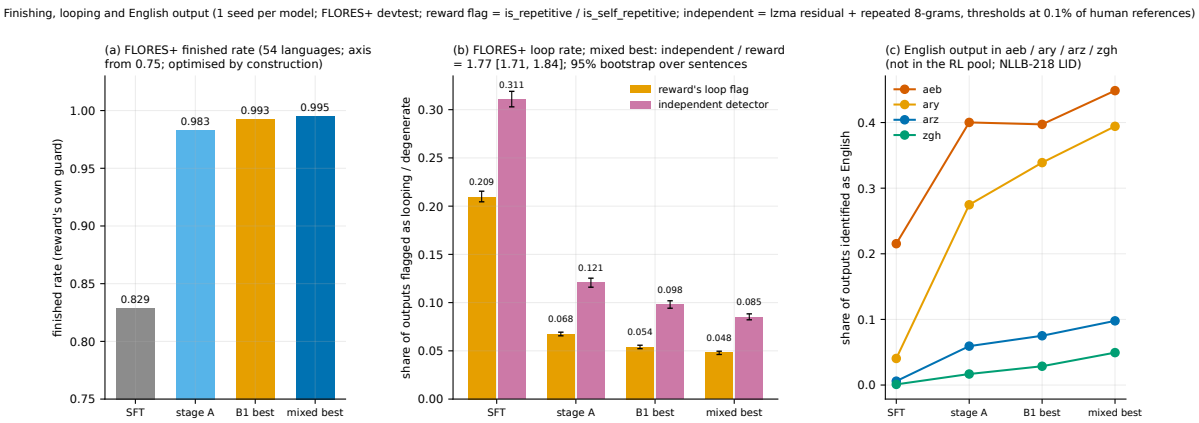


Figure 6: Finishing, repetition and English output on FLORES+ (language averages, one run per model). (a) Share of outputs that finish (axis from 0.75). (b) The reward's repetition check against the independent detector; error bars are 95% intervals. (c) Share of English outputs in four languages that were not among the RL prompts.

Table 6: AfriScience-MT test, sentence level (six languages, one run per stage). SSA-QE = SSA-COMET-QE, of the reward family for every RL row.

model	chrF	AfriCOMET	SSA-QE
SFT baseline	36.00	0.6064	0.5686
first RL stage	45.06	0.6822	0.6118
RL model (mixed best)	45.40	0.6828	0.6130
science RL without science fine-tuning (E-14)	45.66	0.6850	0.6153
science fine-tuning ($\alpha = 0.5$)	46.63	0.6879	0.6118
science RL final	47.10	0.6932	0.6158
long-document RL final	47.16	0.6936	0.6165

starts a fresh Muon state with a 3% linear warm-up. SFT-2 runs in six shards of about 4,800 to 5,300 steps, with tail-language replay from shard 2. Executed stage B: 31,353 steps, 4.109 B tokens, 5.14 s per step on an A100 (44.8 h); 32,853 steps in all with SFT-1. Final validation loss: macro 1.3412, micro 1.2846 over 78 languages (end of SFT-1: 1.7847). The macro rises from 1.78 to 1.98 at the end of shard 1 because new languages enter the validation set at stage 2.

B.2 RL: optimiser, stages, prompts, reward and features

GRPO as used. One on-policy update per sampled batch; a prompt gets a group of 16 completions (15 sampled plus the greedy output; in guided groups 14 sampled, the guide and the greedy output); the advantage is the group-standardised reward; the loss is the token-level policy-gradient term plus a k3 KL penalty to a reference model (coefficient 0.05 in stages A to mixed; the reference is the start model of each stage, i.e. the SFT final for stage A and the previous stage's checkpoint afterwards, because every stage restarts with a fresh reference and a fresh optimiser) (Shao et al., 2024; Schulman, 2020). Optimiser: Muon (momentum 0.95, 5 Newton-Schulz steps), **learning rate 2e-6** with 10 warm-up steps, weight decay 0, gradient clipping 1.0; temperature 1.0, top-p 1.0; at most 384 new tokens in the short-window bin (b64) and 1,280 in b256 and the mixed stage; seed 1234. A KL guard cuts the learning rate when the median KL exceeds a threshold (0.15 in stage A, 0.25 later). Prompts per step: 64 in stage A, 48 in B1, 42 in mixed.

The science SFT before the science RL used AdamW at 3e-6 (Section 5.5). Dynamic-sampling, std-floor and reference-domain settings differ by stage as well (Appendix D).

Stages. Stage A (400 steps, short English windows, bin b64), B1 (135 steps run, b256; best checkpoint step 50 by dev chrF++), mixed (60 steps, b64 and b256), then a continued-RL run on eight weak languages (85 steps), a science SFT with weight interpolation, a science RL and a long-document RL (Appendix D). B1 started from stage A's final model (step 400), mixed from B1's best checkpoint (step 50), the weak-group run from the mixed best, the science SFT from the mixed best. The dev set has **720 sentences = 18 FLORES+ dev languages** $\times$ **40 sentences** (afr ewe fon hau ibo kin nya orm sna som sot swh tsn twi wol xho yor zul). B1's step-135 weights (Hub branch `stage-b1-final`) were evaluated after the fact (E-20,

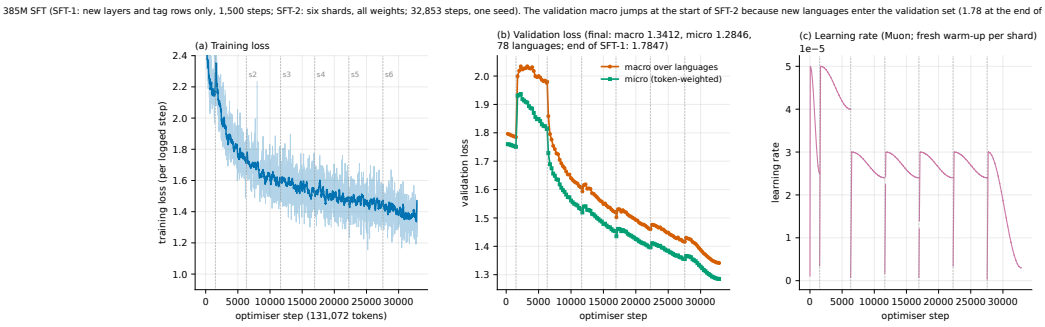


Figure 7: Training of the 385M SFT (one seed): (a) training loss per logged step (thin) and 20-point mean (thick), (b) validation loss (macro over languages and token-weighted micro) and (c) learning rate over the 32,853 optimiser steps of SFT-1 (1,500 steps, new layers and tag rows only) and the six SFT-2 shards (dotted lines); the validation macro jumps at the start of SFT-2 because new languages enter the validation set, and the learning-rate panel shows the per-shard schedule of Section 3. Final validation loss: macro 1.3412, micro 1.2846 over 78 languages. Source: `reports/data/e23_hygiene_2026-11-04/sft_385m_curve.csv`.

Table 7: RL hyper-parameters by stage.

stage	learning rate	KL coefficient (k3)	group size	guide member	steps (cap) × prompts
A, B1, mixed	2e-6 (10 warm-up steps)	0.05	16 (15 + greedy)	beam/MBR, fraction 0.25 (from step 41 in A)	400 × 64; 150 (135 run) × 48; 60 × 42
science RL	1.0e-6	0.1	16	beam/MBR, fraction 0.25	150 in the yaml, cap extended to 190 by the extension rule at step 120; executed 160 (stopped at step 150 by dev plateau, then 10 wind-down steps; `summary.json`, Appendix D) × 42
long-document RL	1.0e-6	0.1	8 (7 + greedy)	off	80 × 32 (51% long prompts)

Section 5.4) and equal B1 best within noise.

Prompts. English windows of consecutive sentences from FineWeb-Edu and Wikipedia (70%) and from news (BBC, Wikinews, Global Voices, and Wikivoyage; 30%), filtered against the protected index; 81 languages with higher draw weights for the 29 weakest. The stage-A news pool contained Wikivoyage (`darkpredator/stealth-wikivoyage-en`; 188 prompts drawn in steps 1-400) and an unlisted source `SetFit/bbc-news` (37 pool rows, none drawn).

Reward. Per completion: hard failures (missing or wrong language tag, unfinished or text after the end token, copy of the source, degenerate repetition, wrong language by GlotLID (Kargaran et al., 2023) with a 15-character minimum, length outside the human window) receive the floor reward −2. Otherwise the content signal is a reference-free quality-estimation ensemble of `masakhane/africomet-qe-stl-1.1` (Wang et al., 2024) (r1) and `McGill-NLP/ssa-comet-mtl` run without a reference (Li et al., 2025) (r2), each shaped per language onto a monotone curve calibrated on human references (held-out calibration split of 261 ids, never dev, devtest or test). **The ensemble is the mean of the two shaped rewards if |r1 - r2| ≤ 1.0 and the minimum otherwise**; it exists for 42 languages (41 with an ensemble block; Dyula has none). For languages without a validated QE model the content signal is the consensus (mean sentence chrF++ against the other valid group members, at least 3 valid members, blend 0.5); for five weak languages (kea kbp nus bam kab) an NLLB-3.3B pseudo-reference chrF++ term is added with weight 0.25 (max 0.35), reward only. Penalties are penalties only (never a bonus), as executed in stages A, B1 and mixed: length (weight 1.0 per log-unit outside the language's human length-ratio window, widened by 15%), word ratio (0.5 per log-unit of the words ratio above the human 99th percentile, slack 15%), lexical (1.0 per unit of deficit in numbers, names and content words that are not in the source), diacritics (1.0), and the constraint anchors (boilerplate, copy with cap 2.0, entity recall 0.5, function words 0.3, padding, repetition and a lexicon guard; v1 for consensus and blend languages, v2 for all three content modes; a saturated stuffing deficit at 0.9 is a hard failure); the fluency and science-anchor penalties were off; hard-failure classes also include an empty output. The reward range is [−2, 2.25]. The reward is computed per sentence chunk and aggregated by source length, with failed chunks at −2 (long-document reward `longdoc_v2`, applied to short bins too).

Four features added at step 41 of stage A. Stage A ran steps 1-40 without them; at the step-40 checkpoint the trainer was stopped and resumed from `last/` with all four switched on. (1) Dynamic

Table 8: chrF++ of the 385M SFT against A0 (language macro, all rows; 95% paired bootstrap). Language counts differ between benchmarks.

benchmark (languages)	chrF++ 385M SFT	chrF++ A0	delta [95% CI]
FLORES+ devtest (46)	25.52	20.20	+5.31 [+5.13, +5.50]
NTREX-128 (24)	29.25	23.01	+6.24 [+6.02, +6.45]
MAFAND-MT test (15)	35.77	25.56	+10.21 [+9.90, +10.54]
SmolSent (45)	22.53	17.18	+5.36 [+5.14, +5.60]
SALT (5)	35.49	25.13	+10.36 [+9.29, +11.44]
GlotStoryBook (8)	20.22	14.62	+5.59 [+4.92, +6.21]
IbomMT (2)	12.04	9.87	+2.16 [+1.58, +2.74]
FLORES-MOZ (1)	21.54	21.08	+0.46 [−0.72, +1.73]

sampling: oversample 1.1 (later 1.0), drop flat groups, refill up to twice (Yu et al., 2025); (2) an advantage standard-deviation floor of 0.02; (3) a beam-guided group member: for 25% of the prompts the beam-4 output (or, for 31 weak languages, an MBR winner over 16 extra samples; skipped on 99% of B1 steps for cost) replaces the last sampled member, with ratio 1 and the guides' advantage share capped at 0.25; (4) a reference-anchored term on 15% of the prompts (7 languages: amh efi hau lug nso yor zul, human pairs from AfriScience-MT train and IbomMT train): r = 0.65 r_existing + 0.35 consensus_value(chrF++ against the human reference). In stage A the std floor never bound (group-std p10 0.128 against 0.02) and dynamic sampling never acted (flat fraction 0); the beam guide cost a median 26.6% of step time and was above the mean of the other members in 83% of groups. The stage-A log cannot separate the four features, so we ablated them at 100 steps (E-11, Appendix G): together they are worth a small gain (+0.32 chrF++), mostly the reference-anchored term, the beam guide is not shown to be worth its 25% step time, and none of them causes the KL rise; we describe them as design choices, not as contributions. The archived stage-A `config.json` is the resumed (post step 40) configuration, so the values of steps 1-40 are known only from the stage report.

Learning rate in stage A. The KL guard cut the learning rate at steps 226 (x0.5), 236 (x0.25) and 247 (x0.125) and it never recovered: steps 247-400 ran at 2.5e-7 instead of 2e-6. The dev curve was flat from about step 100, before the first cut.

Monitors and stops. From the mixed stage on, the held-out reference-based `masakhane/africomet-stl-1.1` is scored at every dev evaluation (never a reward) and enters a hacking alarm (QE rising while dev chrF++ falls by more than 0.75 and AfriCOMET by more than 0.03) and a combined dev-plateau stop (two dev evaluations, 50 steps, without a gain of 0.15 chrF++ or 0.004 AfriCOMET).

Cost. Stage A 15,920 s for 400 steps (39.8 s per step), B1 10,479 s for 135 steps, mixed 4,786 s for 60 steps; the chain is 8.66 A100-hours.

C SFT against A0 and the earlier 270M work (details)

On FLORES+ 38 of 46 languages are significantly better, 6 are not different (kab orm lua lug seh nso) and 2 are worse (kea −17.56 [−18.96, −16.10], dyu −2.33 [−3.06, −1.60]). Held-out AfriCOMET is significantly better for every language that has it (FLORES+ 0.554 against 0.442). The comparison confounds depth with the additional SFT (stage B, 40 added languages, NLLB-mined/Aya/Tumbuka data). Kabuverdianu falls from 36.47 to 18.91 with finished rate 0.98 → 0.83 and 82% NLLB-mined stage-B tokens; its cause is untested.

On exactly the sentences the old 270M models had not seen, the 385M SFT is behind the old SFT on seven of eight benchmarks (FLORES+ 22.34 against 25.06, NTREX 27.35 against 31.46, MAFAND 34.80 against 38.18, SmolSent 17.03 against 19.97, SALT 30.60 against 33.58, IbomMT 9.49 against 13.96, FLORES-MOZ 19.72 against 22.96; GlotStoryBook 20.27 against 19.85). The block of Sotho-Tswana, Tsonga, hau, run, kin, ssw, orm, kab regresses most (lug −15.3, nso −15.3, tso −14.8, sot −11.7). The regression already exists with A0, which has no NLLB-mined rows; the link to NLLB share is inconclusive (Appendix M).

Same harness (E-13). The earlier 270M work's 270M SFT and RL-final (rl2c) were generated again with the 385M harness (greedy, 768 tokens, vLLM bf16) and compared with the 385M SFT and RL (mixed best) on rows that are clean for both generations, with a paired bootstrap (1,000 resamples, 95%

Table 9: chrF++ of the earlier 270M work's 270M SFT and RL-final (rl2c) re-run in the 385M harness (E-13), against the 385M SFT and RL (mixed best), on rows clean for both generations; language macro.

benchmark (languages)	old 270M SFT	385M SFT	385M SFT - old SFT	old 270M RL (rl2c)	385M RL	385M RL - old RL
FLORES+ devtest (45)	24.69	22.34	−2.35 [−3.25, −1.38]	31.68	32.03	+0.35 [+0.08, +0.62]
NTREX (24)	31.46	27.33	−4.13 [−4.40, −3.85]	35.52	35.55	+0.02 [−0.07, +0.11]
MAFAND (15)	38.26	34.71	−3.54 [−4.22, −2.75]	41.80	40.25	−1.55 [−1.72, −1.39]

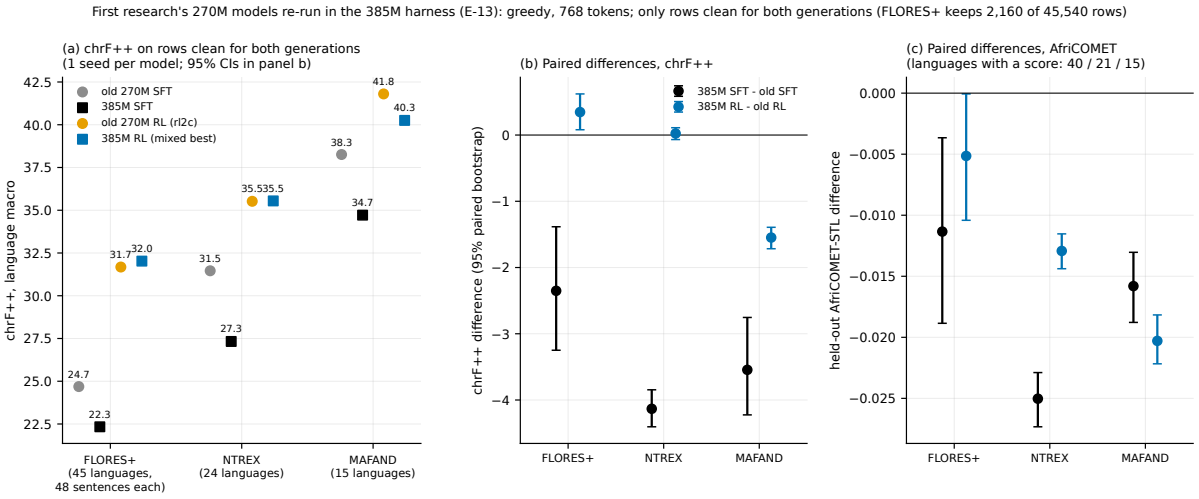


Figure 8: The earlier 270M work's 270M models re-run in the 385M harness (E-13): (a) language-macro chrF++ of the old 270M SFT and RL-final (rl2c) and the 385M SFT and RL (mixed best) on FLORES+ devtest (45 languages, 48 sentences each), NTREX (24) and MAFAND (15), rows clean for both generations only; (b) paired differences with 95% bootstrap over sentences (Poisson weights; 1,000 resamples); (c) the same for held-out AfriCOMET-STL (languages with a score: 40 / 21 / 15). One seed per model. Source: `reports/data/e13_old270m_same_harness_2026-11-04/evidence_e13.json`.

CI).

The 385M SFT is behind the old SFT on all three benchmarks and on held-out AfriCOMET (−0.0113 [−0.0189, −0.0037], −0.0250 [−0.0273, −0.0229], −0.0158 [−0.0188, −0.0130]); after RL the 385M model is level with the old RL on FLORES+ and NTREX in chrF++ and 1.55 chrF++ below it on MAFAND, and lower on AfriCOMET on all three (−0.0051 [−0.0104, −0.0001], −0.0129 [−0.0144, −0.0115], −0.0203 [−0.0222, −0.0182]). The rerun reproduces the stored old-model numbers to within 0.4 chrF++ (old SFT FLORES+ 24.69 against 25.06). Caveats: the clean-for-both filter keeps 2,160 of 45,540 FLORES+ rows (4.7%; 48 sentences per language) and 28,680 of 47,928 NTREX rows, so these 385M numbers are on a different row set from those of Section 5.2; the generations differ in data, steps, tokenizer layout, reward and code state; one seed per model; SmolSent, SALT, GlotStoryBook, IbomMT and FLORES-MOZ were not re-run for the old models (the earlier "7 of 8 benchmarks" statement is not re-measured for those five).

C.1 Weak and added languages

After SFT, 37 of 74 scored languages are weak (chrF++ < 14 or finished rate < 0.9), 16 of them by chrF++: shi 0.2, ibb 2.3, zgh 2.3, dag 2.4, aeb 2.8, taq 2.8, dik 3.8, knc 3.8, arz 4.6, ary 4.6, dyu 9.9, tir 10.5, cjk 10.7, orm 12.1, tiv 13.3, nus 13.5. Of the nine languages that the SFT report lists as added (the eight on FLORES+ that are among the 40 registry rows appended after A0, plus seh, which the registry counts among the first 85 rows), mfe (28.5) and seh (21.7) work, cjk (10.7) is weak, and arz, ary, dik, aeb, taq and zgh fail (English or Modern Standard Arabic output, token loops). After RL (mixed best): mfe 37.93, cjk 23.02, dik 14.06, taq 11.06, arz 9.18, ary 6.88, aeb 4.12, zgh 4.07. Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri (knc) are not usable after RL (chrF++ 2 to 14; loops in 11% to 56% of FLORES+ outputs by the reward's flag, and an English-output share of up to 0.5 in aeb and ary, Section 5.2).

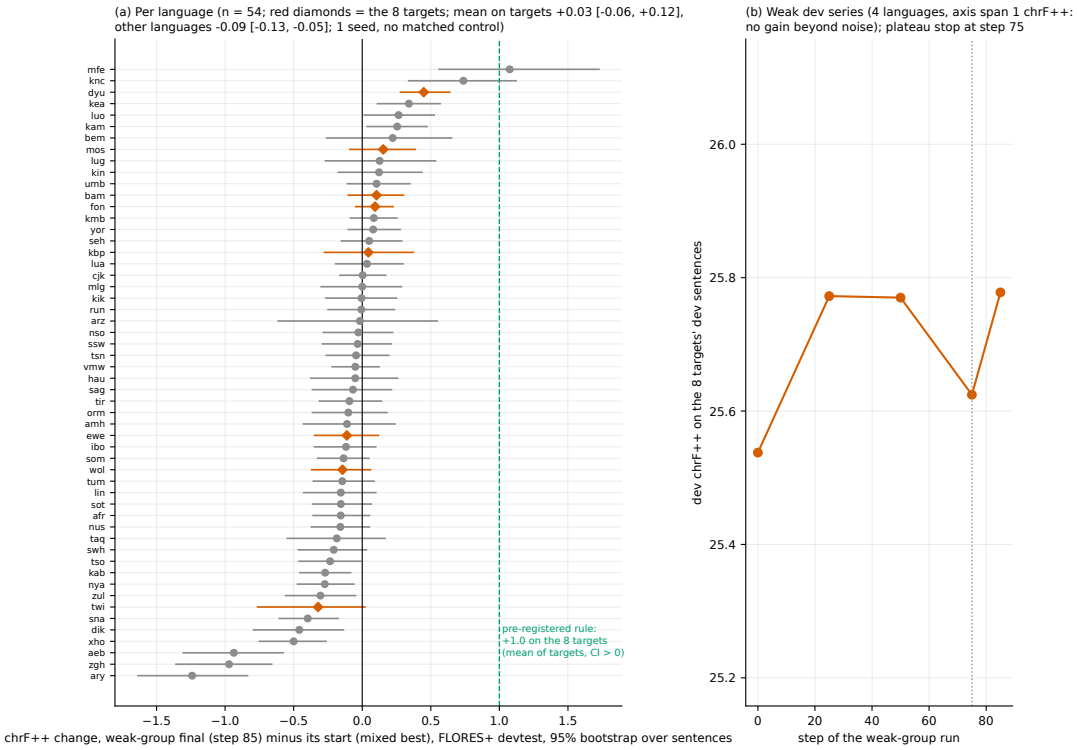


Figure 9: (a) Continued RL on eight weak West-African languages (mos fon kbp bam wol ewe twi dyu; 85 steps, 2.1 GPU-hours): chrF++ change of the final (step 85) against its start, the mixed best, per FLORES+ language (n = 54; 1,012 sentences each; 95% bootstrap over sentences; one seed, no matched control). Red diamonds are the targets: mean +0.03 [−0.06, +0.12] against the pre-registered +1.0; the 46 other languages move by −0.09 [−0.13, −0.05]. (b) The run's own dev series on the targets' dev sentences (axis span 1 chrF++; dev evaluation noise is sd 0.09 to 0.10). Sources: `reports/data/eval_weak_group_2026-11-03/scores/bootstrap_by_language.csv`, `research_archive_2026-11-03/clean_rerun_misc/runs_385m_weak/mixed/metrics.jsonl`.

D RL chain: stage summary, B1 final and training records

Stage A accounts for 94% of the SFT-to-B1 gain on FLORES+ (7.96 of 8.50) and 96% on NTREX; it also accounts for most of the improvement in finishing and looping (FLORES+ finished 0.829 → 0.983, loop 0.209 → 0.068; final 0.995, 0.048; Section 5.2). We did not split the part beyond finishing by stage. The stage-A dev chrF++ (720 sentences) moves 36.81 → 39.04 (step 50) → 40.14 (100) → 40.47 (200) → 40.50 (400), with +3.3 of +3.7 in the first 100 steps; the plateau begins before the three KL-guard cuts (steps 226, 236, 247). The later stages differ from stage A in more than step count (bin, reference domain, guides), and the later steps are small. Two extra seeds of the stage-A recipe (E-04, 100 steps, 20 languages, standard harness; Appendix G) differ by 0.06 chrF++ [−0.02, +0.14] (0.0008 AfriCOMET). On those 20 languages the stored B1 best is +0.17 [+0.12, +0.21] chrF++ over stage A, about three times the seed difference, while the mixed best is +0.07 [+0.02, +0.11] over B1 best, the size of the seed difference: **we do not call the mixed step a gain**; stage A to mixed is +0.23 [+0.18, +0.29]. These are indications from one pair of seeds at step 100, not tests of the later stages.

B1 best against B1 final (E-20). The B1 final weights (step 135; Hub branch `stage-b1-final`) were evaluated on the same 122,585 rows as every other stage (greedy, 768 tokens, language macro, 95% paired bootstrap, one seed).

Held-out AfriCOMET, B1 final minus B1 best, all cells: +0.0004 [+0.0000, +0.0008]; FLORES+ finished / loop 0.993 / 0.054 (B1 best) and 0.996 / 0.052 (B1 final). The 85 steps after step 50 changed nothing measurable, so the choice between B1 best and final does not matter for the numbers of this paper; 11 of 93 cells are significantly worse and 8 significantly better, by 0.1 to 0.3 chrF++, which is a rough noise floor for the stage-to-stage steps. (A table, not a figure, because the content is three differences.)

Table 10: Stage chain summary (not all rows are FLORES+ stages).

stage	steps run	cost	result on FLORES+ chrF++ (vs previous stage)
SFT-1 / SFT-2 (A0 → 385M)	1,500 + 31,353	44.8 h for stage B (A100)	22.85 (SFT, 54 languages)
continued-SFT control (E-03; not an RL stage)	6,000	8.4 h (5.05 s/step), RL chain 8.66 h	FLORES+ 25.38 (+2.53 over the SFT [+2.35, +2.71]); 93 cells 29.97 against 34.55 for the mixed best
RL stage A	400	15,920 s (4.42 h)	30.81 (+7.96)
RL B1	135 (best step 50)	10,479 s	31.35 (+0.54)
RL mixed	60	4,786 s	31.66 (+0.31)
weak-group RL (8 languages)	85 (plateau stop at 75)	2.1 GPU-h	targets +0.03 [−0.06, +0.12]; other languages −0.09 [−0.13, −0.05]
science SFT	283 (65,536 tokens)	17 min	AfriScience chrF +1.2; SSA-COMET-QE flat
science RL	160 steps executed (cap 150 in the yaml, extended to 190 at step 120; dev plateau at step 150 plus 10 wind-down steps; `summary.json` stop reason `dev_plateau`; best at step 75)	not stated in the evaluation report	AfriScience test (6 languages), final vs start: chrF +0.46 [+0.27, +0.65], held-out AfriCOMET +0.0053 [+0.0036, +0.0071], SSA-COMET-QE +0.0041 (reward family); NTREX +0.42, MAFAND +0.10 chrF++; no document-level gain (not a FLORES+ stage)
science RL without the science SFT (E-14)	160 (150 + 10 wind-down; one OOM restart from step 20; final only)	about 5 h wall-clock (15:18 to 20:17 UTC; 14,395 s in the last process)	AfriScience test chrF 45.66 (start 45.40; science RL from the science SFT 47.10); 93-cell chrF++ +0.13 over the mixed best (not a FLORES+ stage)
long-document RL	80 (best at step 50; no extension)	7.7 h wall-clock, of which about 4.5 h a crash loop (Appendix S)	whole-document finished +0.125, loop −0.113, chrF +3.4; chunked and sentence level unchanged (not a FLORES+ stage)

Table 11: B1 final (step 135) against B1 best (step 50) and neighbouring stages; chrF++, language macro, 95% paired bootstrap.

comparison	FLORES+ chrF++	NTREX chrF++	MAFAND chrF++	all 93 cells chrF++
B1 final minus B1 best	+0.04 [+0.00, +0.08]	−0.00 [−0.05, +0.05]	−0.09 [−0.15, −0.01]	+0.01 [−0.02, +0.04]
B1 final minus stage A	+0.58 [+0.53, +0.64]	+0.30 [+0.24, +0.36]	−0.09 [−0.17, +0.00]	
mixed best minus B1 final	+0.26 [+0.22, +0.30]	+0.19 [+0.14, +0.23]	+0.05 [−0.01, +0.12]	

E RL against SFT and the control: details

The loop detector in this table is the reward's own guard (`is_repetitive`, `is_self_repetitive`); on the human references it fires on 0.0001 of FLORES+ rows. 95% of the RL model's flagged FLORES+ rows come from seven near-failed languages (taq knc zgh dik aeb arz ary). Section 5.2 tests this flag against a detector that is not the reward's.

Cutting the SFT's looping outputs at the loop onset (no change to the translation) raises the SFT from 26.59 to 30.49 chrF++, i.e. **49% of the gain is recovered by cutting loops**; 67 to 70% of the chrF++ gain is attributable to rows where either model failed; about 41% of the AfriCOMET gain is finishing. What remains is +2.4 to +4.1 chrF++ and +0.037 AfriCOMET, significant in 50 of 54 FLORES+ languages on clean rows. The three estimates favour different models (the paired subset keeps only sentences the SFT can handle; the cut-at-loop rule is a heuristic) and agree in sign and rough size. The cut-at-loop rule is the reward family's loop rule; it was not redone with the independent detector of Section 5.2. The decode-time guard control (E-01, below) answers the question this raises: a frequency penalty of 0.10 on the SFT recovers the same 49% of the chrF++ gain (29% of the AfriCOMET gain) without any RL, so the finishing share is a property of unguarded greedy decoding that the RL also repairs. Caveats: the guard setting was chosen on the FLORES+ dev held-out half by a rule fixed in advance and its optimum (0.10) lay at the edge of the grid, so the control may be under-tuned; a post-hoc dev-only extension to stronger penalties (SFT dev chrF++ 37.36 at 0.10, 37.97 at 0.20, 38.00 at 0.30, 38.25 at 0.50) was still rising and no full evaluation with a stronger penalty was run; one seed (`reports/e01_loop_guard_2026-11-04.md`, `reports/overnight_log_2026-11-04.md` 09:00 row).

Language-macro chrF++ per benchmark, SFT / control / mixed best: FLORES+ 22.85 / 25.38 / 31.66, NTREX 29.25 / 33.74 / 37.17, MAFAND 35.77 / 40.44 / 40.78; the control gains +2.53 [+2.35, +2.71], +4.49 [+4.29, +4.70] and +4.67 [+4.39, +4.92] over the SFT against +8.81, +7.91 and +5.01 for the RL; over the 93 cells it gains +3.38 against +7.96 (42.5% of the SFT-to-RL gap, arithmetic of the macros). Held-out AfriCOMET: RL minus control is +0.0373 on FLORES+, +0.0051 on NTREX and −0.0141 on

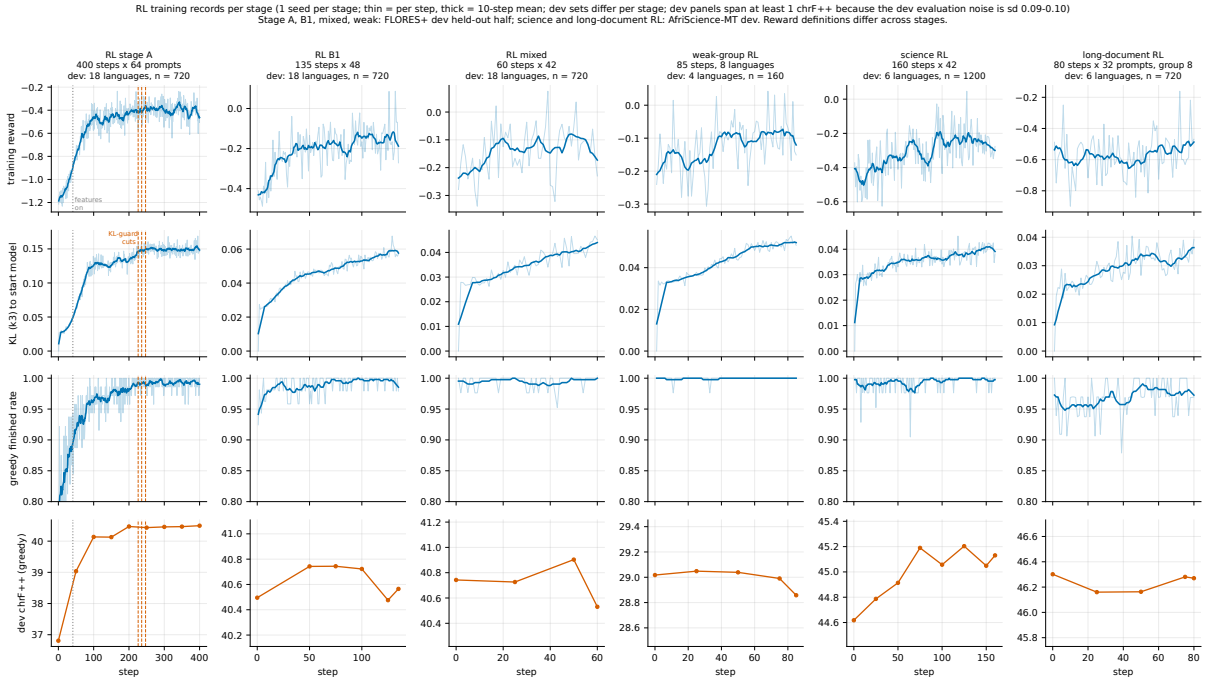


Figure 10: Training records of every RL stage from the run logs (one seed per stage): training reward, KL (k3) to the stage's start model, finished rate of the greedy group member, and dev chrF++ against the step (thin lines per step, thick lines 10-step means). Stage A: dotted line at step 41 (four features on), dashed lines at the three KL-guard cuts (226, 236, 247). The dev sets differ by stage (FLORES+ dev held-out half for stage A, B1, mixed and weak; AfriScience-MT dev for science and long-document RL; the number of languages and sentences is in each title) and the reward definitions differ, so columns are not comparable; dev panels span at least 1 chrF++ because the dev evaluation noise is sd 0.09 to 0.10. The long-document RL log shows the 80 planned steps; step 25 was evaluated repeatedly after the crash restarts of Appendix S and the last evaluation is shown (steps 21 to 25 were repeated by the restarts). Sources: `research_archive_2026-11-03/runs_385m/{A,B1,mixed}/metrics.jsonl`, `clean_rerun_misc/runs_385m_{weak/mixed,sci/final,long/final}/metrics.jsonl`.

Table 12: Finished rate / loop rate (reward's flag) / hypothesis-to-reference length, clean rows, language macro.

system	FLORES+ finished / loop / hyp-to-ref length	NTREX	MAFAND
SFT	0.829 / 0.209 / 3.03	0.890 / 0.122 / 2.35	0.963 / 0.043 / 1.48
RL stage A	0.983 / 0.068 / 1.33	0.996 / 0.016 / 1.07	0.999 / 0.003 / 1.00
RL B1 best	0.993 / 0.054 / 1.17	0.999 / 0.009 / 1.02	1.000 / 0.001 / 0.99
RL mixed best	0.995 / 0.048 / 1.14	0.999 / 0.006 / 1.02	1.000 / 0.001 / 0.98

MAFAND (the last favours the control). Finished and loop rates (the reward's flag) on FLORES+, SFT / control / mixed best: 0.829 / 0.854 / 0.995 and 0.209 / 0.178 / 0.048; by panel (FLORES+ finished): the 19 TranslatePsy languages 0.931 / 0.993 / 1.000, the other languages 0.774 / 0.779 / 0.993. The independent degeneration detector gives the same picture (FLORES+ flag 0.311 for the SFT, 0.268 for the control, 0.085 for the mixed best). **The control repairs finishing and looping only on the languages it was trained on**, where it comes close to the RL model; on the other languages it changes almost nothing (+0.15 chrF++ over the SFT on FLORES+ [−0.03, +0.32], 35 languages).

By the rule as registered the verdict is "inconclusive", because the condition on the 19 TranslatePsy languages is not met (d = −0.03); the other three are. We did not change the rule. Reading the panels: where a teacher-translated corpus exists continued SFT matches RL in chrF++ and largely fixes finishing and looping, and the control is ahead of RL on the neural metrics there; RL's advantage is that it improves every language from its own samples and a QE reward, including those for which no such corpus was used (+9.29 on 46 cells). A reader who takes the overall endpoint alone gets "RL is better" (+4.58 [+4.45, +4.70], above the +1.0 margin, AfriCOMET above 0); a reader who takes the 19 languages alone gets "comparable" in chrF++ (interval inside ±1.0) but not by the registered rule (the AfriCOMET interval excludes 0 in favour of the control); neither is the registered verdict. On AfriScience-MT test sentences (6 languages) the control scores chrF 44.71 against 45.40 for the mixed best and 36.00 for the SFT, and is ahead of the RL on SSA-COMET-QE (0.6214 against 0.6130) and AfriCOMET (0.6947 against 0.6828), the metrics in the family that filtered TranslatePsy; nso, the one of the six languages without TranslatePsy

Table 13: RL minus SFT over the 93 cells under four ways of removing the finishing effect (clean rows; 95% CI).

row set	chrF++ RL - SFT [95% CI]	AfriCOMET RL - SFT [95% CI] (76 cells)
all clean rows	+7.96 [+7.82, +8.12]	+0.0631 [+0.0620, +0.0643]
SFT output cut at its first loop (RL as is)	+4.06 [+3.97, +4.15]	not scorable
rows where the SFT finished and did not loop	+2.41 [+2.33, +2.49]	+0.0366 [+0.0357, +0.0374]
rows where both finished and neither loops	+2.61 [+2.54, +2.70]	+0.0370 [+0.0362, +0.0379]
rows where the SFT failed (21% of FLORES+ rows)	+24.57 [+24.26, +24.90] (SFT 5.95, RL 30.53)	+0.3342 [+0.3295, +0.3391]

Table 14: The pre-registered decision rule applied to mixed best against the 6,000-step control.

registered condition	value	met
lower 95% bound of d (chrF++, 93 cells) $> +1.0$	$+4.58$ $[+4.45, +4.70]$	yes
lower bound of the AfriCOMET difference > 0	$+0.0183$ $[+0.0172, +0.0192]$	yes
$d > 0$ on the 19 TranslatePsy languages	-0.03 $[-0.13, +0.09]$	**no**
$d > 0$ on the other languages	$+9.29$ $[+9.09, +9.49]$	yes

data, stays at the SFT level for the control (41.27 $\rightarrow$ 40.64) while the RL gains 10.9.

Caveats. One seed, **one learning rate** (the probe's best of three) and **one evaluated budget point**: the stage-A-matched (3,100 steps) and mixed-best-matched (4,800 steps) comparisons are untested, and a different mix, schedule or longer run could change the control. Differences below about 0.5 chrF++ are inside the unmeasured seed variation. The control is not information-matched: it receives NLLB-3.3B translations (a teacher that scores 38.80 against 26.30 for the SFT on 43 languages, Section 5.4) for 19 languages that the RL never sees, and the TranslatePsy rows were filtered with AfriCOMET, SSA-COMET and MetricX, so those neural metrics favour the control; chrF and chrF++ are the evidence. The control's loop count is undercounted by the reward's flag too (independent / reward flag 1.51 on FLORES+). TranslatePsy is CC-BY-NC-4.0 with NLLB-derived targets and was used only for this control; nothing of it is in any released model, and the control checkpoint is not released (Appendix R).

F Independent degeneration audit: details

The reward's loop flag is the guard the RL was optimised against, so a loop type it misses would be undercounted for the RL. We re-measured with a detector that shares the idea (repetition) but none of the code or constants: the lzma-compressed size of the output relative to what human references of the same benchmark, language and length would give, the share of repeated character 8-grams and the share of repeated word 3-grams, each threshold set so that 0.1% of the 122,585 human references fire it. Rows are all rows of FLORES+ (54 languages), NTREX (24) and MAFAND (15) for each system, no clean-row filter.

- **The reward's flag is a lower bound.** On mixed-best FLORES+ the independent detector flags 1.77 [1.71, 1.84] times as many outputs as the reward's flag (NTREX 2.25 [1.95, 2.59], MAFAND 4.42 [3.14, 7.29] on tiny base rates). At a ten times stricter threshold (0.01% of references) the FLORES+ rates are 0.0586 against 0.0480 (ratio 1.22), so the excess is not an artefact of a loose threshold. The undercount is present for the SFT too (ratio 1.48), so it enters the RL-versus-SFT comparison only through 1.48 against 1.77; the reduction of the loop rate from SFT to RL is 3.7x with the independent detector and 4.4x with the reward's.
- **The extra rows are finished partial repetitions and cost little.** Of 122,585 mixed-best rows, 2,932 are flagged by both, 14 by the reward only and 2,488 by the independent detector only. The independent-only rows are 99.9% finished, have a median of 26 words and 48% of them are in the seven near-failed languages. Hard failures are caught by both (of 278 unfinished rows 2 escape the reward's flag and none the independent one). If the independent-only rows scored the average of the unflagged rows of their language, the macro chrF++ of the mixed best would rise by 0.19 (FLORES+), 0.02 (NTREX) and 0.015 (MAFAND), against 0.46, 0.28 and 0.23 for the SFT. The difference between SFT and RL in this loss is 0.26, 0.26 and 0.22 chrF++, i.e. 3.0%, 3.3% and 4.4% of the RL gains (+8.81, +7.91, +5.01)

Table 15: Loop flag of the reward against the independent detector, language macro (95% bootstrap in brackets).

system	FLORES+ reward flag	FLORES+ independent flag	NTREX reward / independent	MAFAND reward / independent
SFT	0.2095	0.3107 [0.3029, 0.3190]	0.1218 / 0.1662	0.0431 / 0.0754
RL stage A	0.0675	0.1207 [0.1160, 0.1253]	0.0165 / 0.0309	0.0026 / 0.0092
RL B1 best	0.0541	0.0980 [0.0942, 0.1019]	0.0092 / 0.0190	0.0013 / 0.0056
RL mixed best	0.0480 [0.0462, 0.0496]	0.0851 [0.0822, 0.0883]	0.0064 / 0.0143	0.0009 / 0.0039
human references	0.0001	0.0018	0.0000 / 0.0018	0.0000 / 0.0013

Table 16: Share of FLORES+ outputs identified as English, by system.

language	SFT	stage A	B1 best	mixed best	weak final
aeb	0.215	0.400	0.397	0.449	0.498
ary	0.041	0.275	0.339	0.394	0.488
arz	0.006	0.059	0.075	0.098	0.151
zgh	0.001	0.017	0.029	0.049	0.074
macro, 54 languages	0.0055	0.0146	0.0164	0.0191	0.0235

by arithmetic. **The decomposition above does not change materially.**

- **Length and tails.** Median hypothesis-to-reference character ratio is 1.02 to 1.07 for all systems. The share of FLORES+ outputs longer than twice the reference falls from 0.2005 (SFT) to 0.0281 (mixed best). Of the 2,079 mixed-best rows above 2x, 24.5% (about 0.4% of all rows) are finished and flagged by neither detector: long, non-repetitive continuations that no repetition rule sees. Unterminated tails on FLORES+ fall from 0.1847 to 0.0464 against 0.0162 for human references; NTREX and MAFAND are at or below the reference rate.
- **Language.** An independent language-identification model (NLLB-218, reliable for 40 of the evaluated languages) agrees with the reward's GlotLID: FLORES+ pass rate SFT 0.9487, mixed best 0.9669 (references 0.9915); GlotLID 0.9098 and 0.9553. On mixed-best FLORES+ 0.86% of the rows GlotLID passes fail the NLLB LID, so the language guard's pass rate is not inflated by optimisation. The LID is blind to the weakest languages (none for aeb ary arz mfe nde seh shi ven vmw; unreliable for nine more).

Four languages account for 96% of the macro increase (human references: 0.0004); GlotLID gives the same numbers within 0.005. These languages, with taq, dik and knc, are outside the RL prompt pool, and the model emits the tag of a sibling language (`<arz>` for ary and half of aeb, `<tzm>` for zgh and taq). The reward's copy guard sees only 0.086 of aeb and 0.064 of ary outputs. So part of the "RL loops less" gain in these languages is a change from loops to English output, while their chrF++ stays at 2 to 9.

- **What the audit does not establish.** There is no ground truth for "degenerate"; the detector is validated by the reference control, by reading 25 mixed-best and 12 SFT examples, and by the lower language-matched chrF++. It is blind to non-repetitive failures (hallucination, wrong content) except through the length audit. The NLLB LID is licensed cc-by-nc-4.0 and was used for evaluation only.

The share of outputs identified as English (NLLB-218 top-1, $p \geq 0.5$) on FLORES+ by system:

G Stage-A log analysis and ablation details

The stage-A log cannot separate the four features (E-11 can). Stage A steps 1-40 ran without the four features and steps 41-400 with them. An earlier note said the KL slope "tripled" when the features began (0.00053 per step over steps 2-40, 0.00143 over steps 41-100). Reading the log: both numbers reproduce exactly, **but the factor is 2.0, not 3**: the first segment contains the learning-rate warm-up (steps 1-10); from step 11 the pre-feature slope is 0.00070 [0.00061, 0.00079] and 0.00143/0.00070 = 2.0. The KL was also already accelerating before step 41 (slopes over 15-step windows: 0.00048 for steps 11-25, 0.00101 for 26-40, 0.00158 for 41-55), the curve is smooth through step 41 with no kink, and it saturates by step ~{}90 (flat at 0.124 to 0.129 until step ~{}160). A power-law extrapolation of the pre-feature trend predicts a KL of 0.042 to 0.065 at step 100 against 0.131 observed, so the acceleration is faster than the extrapolated trend, but that is a statement about extrapolating 30 to 40 noisy points, not an identification

Table 17: Ablation arms at 100 steps, chrF++ over 49 cells (20 languages); arm minus the full-feature seed 2 (paired) and minus seed 3 (E-11).

arm	chrF++ (49 cells)	arm minus seed 2	arm minus seed 3	AfriCOMET vs seed 2
(b) full, seed 2	38.51 [38.30, 38.69]	-	-	-
(b) full, seed 3	38.46 [38.26, 38.64]	-	-	-
(a) no features	38.19 [37.99, 38.38]	−0.32 [−0.41, −0.23]	−0.26 [−0.35, −0.18]	−0.0028 [−0.0033, −0.0022]
(c) no reference domain	38.35 [38.16, 38.54]	−0.16 [−0.25, −0.08]	−0.10 [−0.19, −0.02]	−0.0017 [−0.0023, −0.0011]
(d) no beam guide	38.42 [38.23, 38.61]	−0.09 [−0.17, −0.01]	−0.04 [−0.11, +0.04]	−0.0005 [−0.0010, +0.0001]

of a cause.

Nothing else accelerates at step 41. The per-step slopes of the training reward (+0.0065 [0.0051, 0.0079] before, +0.0071 [0.0060, 0.0080] after), of the finished rate, of the greedy loop rate (−0.0020 before, −0.0011 after) and of the hard-fail share (−0.00138 before, −0.00040 after) are the same or steeper before step 41, because the failure rates were high and floor out. An interrupted-time-series fit finds level shifts at step 41 (KL +0.0086 [0.0021, 0.0149], reward +0.093 [0.041, 0.147], greedy loop rate −0.028 [−0.052, −0.004], hard-fail share −0.014 [−0.026, −0.003]), but the reward and the finished and loop rates are not the same quantity before and after step 41 (from step 41, 15% of the prompts carry the reference term and 25% of the groups contain the guide), and the shifts could equally be a resume artefact (checkpoint reload, RNG, a fresh prompt stream). They say "something changed at 41 on top of the trend", not which feature did it. The dev curve is evaluated only at steps 0, 50, ..., 400, so no dev point separates steps 1-40 from 41-50.

Which feature could act at all. Dynamic sampling never acted (flat-group fraction 0, no refill) and the std floor never bound (group-std p10 0.128 against 0.02), so any effect at step 41 could only come from the beam guide, the reference domain or the resume itself; the log could not say which. We therefore ran the ablations (below).

Seed spread (E-04). Two extra runs of the stage-A recipe (seeds 20261105 and 20261106; 100 steps; the executed stage A used seed 1234, so the three runs share the recipe, the SFT start and the 40-step feature-free prefix schedule) were evaluated with the standard harness on 20 languages (yor hau ibo swh amh zul xho kin som wol lug sna twi ewe lin tir orm nso tsn afr; FLORES+, NTREX, MAFAND; 49 cells; greedy, 768 tokens). Gains over the SFT are +5.96 [+5.80, +6.11] and +5.90 [+5.74, +6.05] chrF++ (+0.0462 and +0.0454 AfriCOMET): the two seeds differ by **0.06 chrF++ [−0.02, +0.14]** (0.0008 AfriCOMET; per language they differ significantly in 2 of 20, by at most 0.54). The in-training dev gains of the three runs are +3.33, +3.70 and +3.46 (the dev subset depends on the seed, so only gains over each run's own step 0 are compared). The stage-A gain is therefore reproducible across training seeds at step 100, and per-language differences below about 0.5 chrF++ are not separable from training noise. **This is one pair of seeds at step 100, not a seed study of later stages**; the 400-step stage A is +0.78 [+0.70, +0.86] and +0.84 [+0.75, +0.92] above the 100-step seeds on the same cells.

Ablations (E-11). Arms (a) no features, (c) no reference domain (no reference prompts and no reference-anchored reward) and (d) no beam guide were run for 100 steps (b64 recipe, one run per arm) as forks of seed 2's step-40 state, so arm minus seed 2 is a tight paired comparison and arm minus seed 3 carries the full seed noise; arm (b), the full set, is the two seeds. Pre-registered rule: an effect is real if the paired interval excludes 0, its size exceeds the observed seed difference (0.06) and AfriCOMET does not point the other way.

Removing all four features costs **0.32 chrF++ [0.23, 0.41]** (about five times the seed difference; consistent on the three benchmarks, AfriCOMET and the loop flag), about 5% of the 5.96 chrF++ gain, and it does not show on the in-training dev macro (−0.14 [−0.51, +0.25]). Removing the reference domain costs 0.16 [0.08, 0.25] (real by the rule against seed 2, 2.7 times the seed difference): about half of the feature set's effect. Removing the beam guide costs 0.09 against seed 2 but 0.04 [−0.04, +0.11] against seed 3 with AfriCOMET inside noise: **not separable from seed noise**, for about +25% step time (median step 40.3 s with the guide against 32.4 s without). Dynamic sampling and the std floor never acted and are

Table 18: Earlier 256-token study (SFT only; 20 languages × 200 FLORES+ sentences). Beam rows are not comparable with E-02.

decoding	chrF++	delta vs greedy [95% CI]	spBLEU	delta spBLEU
greedy	35.00	-	13.64	-
beam 4	37.53	+2.53 [+2.18, +2.87]	16.38	+2.73 [+2.39, +3.08]
MBR-16 (+ greedy)	38.29	+3.30 [+3.00, +3.58]	15.30	+1.65 [+1.38, +1.90]

Table 19: E-02: greedy, beam-4 and MBR-16 at 768 new tokens, full table (20 languages, 4,000 sentences per system, 95% paired bootstrap).

system	chrF++	spBLEU	AfriCOMET-STL	finished	loop (independent)	output / reference length
SFT greedy	32.31 [31.43, 33.38]	11.47	0.5513 [0.5403, 0.5621]	0.915	0.147	2.05
SFT beam-4 (stop at 4 ended)	36.02 [35.13, 36.95]	16.49	0.5639 [0.5521, 0.5750]	1.000	0.108	0.96
SFT beam-4 (HF early_stopping=False)	33.26 [32.22, 34.46]	13.07	0.5722 [0.5603, 0.5845]	0.897	0.139	2.15
SFT MBR-16 (metric-matched)	38.27 [37.60, 39.01]	15.25	0.5780 [0.5687, 0.5870]	0.9998	0.016	1.16
RL greedy	39.87 [39.17, 40.62]	17.39	0.6082 [0.5991, 0.6172]	1.000	0.007	1.05
RL beam-4 (stop at 4 ended)	39.10 [38.20, 39.96]	17.83	0.6045 [0.5944, 0.6148]	1.000	0.007	0.98
RL beam-4 (HF early_stopping=False)	40.70 [40.00, 41.42]	18.47	0.6227 [0.6139, 0.6317]	0.9998	0.007	1.03
RL MBR-16 (metric-matched)	40.79 [40.09, 41.54]	17.74	0.6168 [0.6079, 0.6258]	1.000	0.003	1.08

part of arm (a) only. **The KL rise is not caused by the features:** the median KL of steps 81 to 90 is 0.116 without any feature, 0.122 without the reference domain, 0.120 without the guide and 0.123 and 0.120 for the two full runs, and the same holds at steps 51 to 60 (0.075, 0.0745, 0.073, 0.072, 0.073). **The earlier inference that the beam guide drives the KL slope (stage report, ledger M10) is therefore not supported**; the step-41 acceleration seen in stage A is at most a small effect. Arm (e) (KL guard 0.25 against 0.15) was not run by the pre-registered rule, because the guard never fired in any of the 100-step runs (maximum 5-step median KL 0.120 to 0.127 against 0.15). Caveats: one run per arm, a shared 40-step prefix with seed 2, one pair of seeds as the noise yardstick (so the "real" rule is lenient and effects of 0.1 to 0.2 chrF++ are small and suggestive), 100 steps and 20 languages only (the reference domain covers amh efi hau lug nso yor zul, five of them in these 20). **Reading:** the four features are design choices with a small measured benefit (+0.3 chrF++ at 100 steps, mostly the reference-anchored term), not the source of the RL gain, and the beam guide is not shown to be worth its generation time.

H Decoding and engine parity: details

Earlier 256-token study (SFT only; superseded for beam). 20 languages × 200 FLORES+ devtest sentences, 256-token cap. Its beam rows came from a vLLM beam loop that completes a beam only on the tokenizer's `<eos>`, not on `<end_of_turn>`, and ran every instance to the cap, so **its beam numbers are not comparable with the beam-4 of E-02 below** and are kept only as the record of what that study measured; its greedy and MBR rows are not affected by this.

MBR minus beam is +0.76 [+0.49, +1.02] chrF++ and −1.08 [−1.39, −0.79] spBLEU; MBR uses chrF++ as its utility. The study's greedy is cap-protected (greedy on the same sentences is 32.50 at 768 tokens against 35.00 at 256), which is one more reason not to use its beam gain.

Decoders help the SFT: beam-4 +3.71 [+2.78, +4.47] chrF++ and +0.0127 [+0.0043, +0.0201] AfriCOMET, MBR-16 +5.96 [+5.37, +6.49] and +0.0267 [+0.0224, +0.0306] (MBR also helps on spBLEU, +3.78, but its chrF++ gain is largest on the metric it optimises); the HF-rule beam only +0.95 [+0.28, +1.66] because it keeps the SFT's looping outputs (finished 0.897). **For the RL model** MBR-16 helps a little (+0.92 [+0.81, +1.03] chrF++, +0.0087 AfriCOMET), beam-4 with the stop-at-4 rule does not (−0.77 [−1.45, −0.25] chrF++, AfriCOMET inside noise; shorter output, length ratio 0.98), and the HF-rule beam gives +0.83 [+0.67, +1.00] chrF++ and +0.0145 AfriCOMET: the sign of the RL beam result depends on the stopping rule. **Decoding does not replace RL:** RL greedy minus the SFT's best decoder (MBR-16) is +1.60 chrF++ [+1.34, +1.88] and +0.0302 AfriCOMET [+0.0270, +0.0335] (against SFT beam-4: +3.85 [+3.29, +4.53]); MBR costs about 17 times the compute per sentence. **The RL gain is partly the same repair that decoding gives:** RL minus SFT is +7.56 [+6.89, +8.19] chrF++ with greedy, +3.07 [+2.67, +3.46] with beam-4 and +2.52 [+2.26, +2.80] with MBR-16 (AfriCOMET +0.057, +0.041, +0.039), interactions −4.48 [−5.18, −3.79] and −5.04 [−5.57, −4.42]; with the HF-rule beam, which does not repair the SFT, the interaction is −0.12 [−0.85, +0.55]. This is the picture of the guard

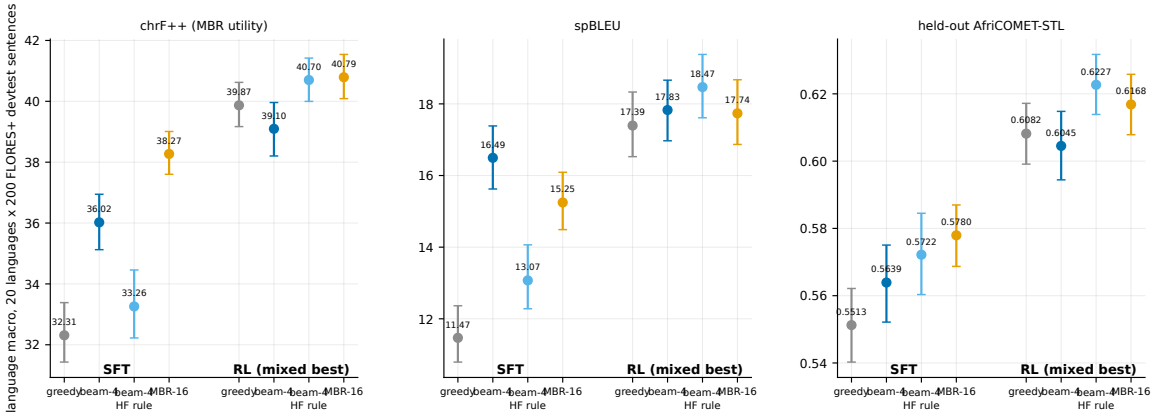


Figure 11: Greedy, beam-4 and MBR-16 at 768 new tokens (E-02) for the SFT and the RL model (mixed best): chrF++, spBLEU and held-out AfriCOMET-STL, language macro over 20 languages × 200 FLORES+ devtest sentences (4,000 per system), 95% paired bootstrap over source sentences; beam-4 stops when 4 hypotheses have ended, the HF rule is early_stopping=False, MBR is metric-matched (utility chrF++). One SFT and one RL checkpoint, vLLM bf16. Not comparable with the earlier 256-token study. Source: `reports/data/e02_beam_mbr_768_2026-11-05/cmp_all.json`.

control (Section 5.2) on a different intervention. The earlier estimate of +5.0 for the beam gain over a 768-token greedy is replaced by the measured +3.71 [+2.78, +4.47] (the interval excludes 5.0). Caveats: 20 languages × 200 sentences, one SFT and one RL checkpoint, vLLM bf16 (absolute values can move by a few tenths of chrF++ with another engine, Appendix Q), the primary beam definition was amended after a smoke test and before any score, and other beam widths, length penalties and the 54-language run were not done.

The decode-time frequency-penalty guard was run on both models over the full evaluation set (E-01, Section 5.2). Engine parity: vLLM bf16 and HF bf16 give identical outputs in 9.6% of 240 sentences (chrF++ 36.22 and 36.80; HF fp32 38.18); in fp32 the two engines agree (98.8% identical for the SFT, 97.1% for the RL model; E-09), so the divergence is bf16 rounding. **Precision does not move the headline:** the SFT in fp32 on the full evaluation set (vLLM, greedy, 768 tokens, 122,585 rows, 214 flagged rows removed; E-09b) scores −0.04 chrF++ [−0.13, +0.07] against bf16 over the 93 cells (FLORES+ −0.16 [−0.29, −0.02], NTREX +0.08 [−0.08, +0.25], MAFAND +0.22 [−0.03, +0.50]; plain chrF −0.08 [−0.19, +0.03]), and the RL gain on FLORES+ is +8.97 [+8.71, +9.20] over the fp32 SFT against +8.81 [+8.59, +9.03] over the bf16 SFT, so the bf16 harness does not overstate the RL gain (the upper end of the all-cell interval bounds any understatement of the SFT at 0.07 chrF++). The +1.81 [+0.59, +3.12] of the 240-sentence set was a small-sample result and the earlier +0.24 [−0.07, +0.55] (4,000 sentences) to +1.81 range for the SFT is superseded. Not done: an fp32 run of the RL model (its sensitivity is measured only on the 240-sentence set, +0.09 [−0.41, +0.59]) and AfriCOMET scoring of the fp32 SFT run (the run was stopped before the scoring finished; the partial scores are not reported).

I NLLB comparison and other systems: details

The RL closes 74% of the gap between the SFT and NLLB-3.3B (12.50 → 3.26; arithmetic of the two differences), and stage A alone gives 8.61 of the 9.24 chrF++ that the mixed best adds over the SFT on these languages (93%). The remaining gap is not a finishing or looping artefact: on these 43 languages the mixed best finishes at least 0.99 and repeats at most 0.02 in every language. By NLLB-3.3B tertile (14 / 14 / 15 languages) the mixed best reaches 91.7% (high, 46.82 against 51.05), 87.7% (mid, 34.57 against 39.42) and 96.8% (tail, 25.92 against 26.78) of NLLB-3.3B; the SFT 75.3%, 63.2% and 60.7%. The mixed best matches or beats NLLB-3.3B on 7 of 43 languages, all in the lowest tertile (yor +5.43, fon +2.95, dyu +2.77, wol +2.48, kam +2.32, kmb +2.22, umb +1.86; the SFT only on fon, 20.17 against 19.30) and is within 1 chrF++ on 9. The largest remaining gaps are amh −11.28, kin −9.74, kea −8.72, run −8.02, ewe −7.03, lug −6.77, tir −6.06 and hau −6.04.

Same sentences. In the post-editor run, NLLB-3.3B and RL stage A drafts were scored on identical 100-sentence FLORES+ subsets (language bootstrap only; sentence hypotheses are not stored). With beam-4 on our side (the beam setup of the post-editor run, not that of E-02) and NLLB greedy on theirs,

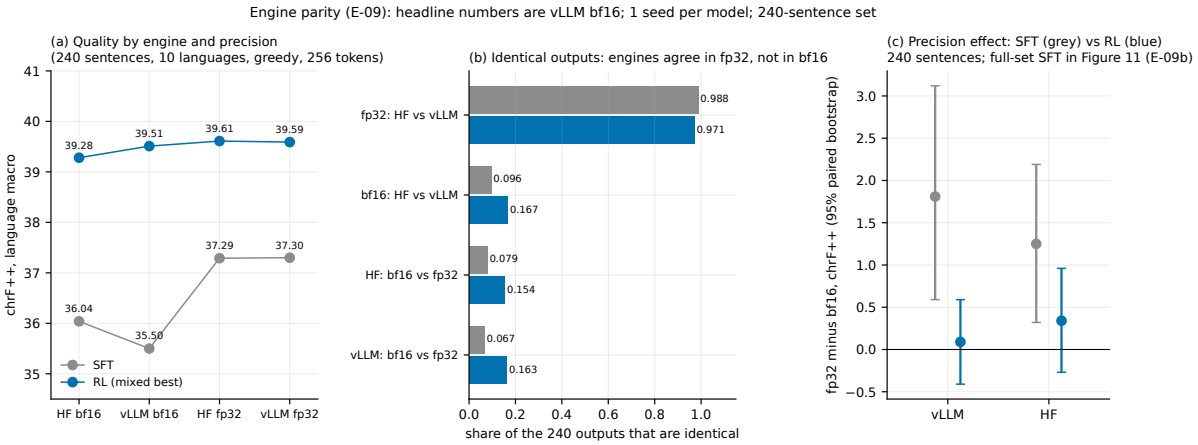


Figure 12: Engine parity (E-09), 240 sentences (10 languages × 24 FLORES+ devtest sentences), greedy, 256 new tokens, SFT and RL (mixed best), one seed. (a) Language-macro chrF++ by engine and precision. (b) Share of identical outputs: vLLM and HF agree in fp32 (0.988, 0.971) and not in bf16 (0.096, 0.167). (c) fp32 minus bf16 chrF++ with 95% paired bootstrap over sentences: negligible for the RL model, +1.81 [+0.59, +3.12] (vLLM) for the SFT on this small set, which the full-set run of Figure 11 does not reproduce. Headline numbers of the paper are vLLM bf16. Sources: `reports/data/e09_engine_parity_2026-11-04/{parity_bootstrap,parity_summary}.json`.

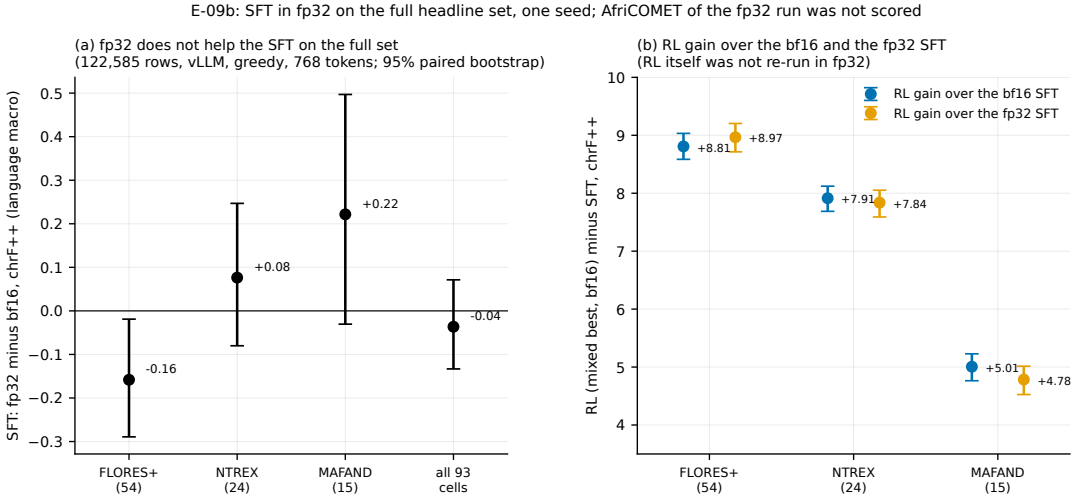


Figure 13: Full-set precision check (E-09b): the SFT generated in fp32 instead of bf16 (vLLM, greedy, 768 new tokens, 122,585 rows, 214 flagged rows removed; one seed). (a) fp32 minus bf16 chrF++ (language macro, 95% paired bootstrap over sentences) on FLORES+ (54 languages), NTREX (24), MAFAND (15) and over all 93 cells (−0.04 [−0.13, +0.07]). (b) RL gain (mixed best, bf16) over the bf16 SFT and over the fp32 SFT. The RL model was not re-run in fp32 and the AfriCOMET of the fp32 run was not scored. Source: `reports/data/e09b_fp32_2026-11-04/evidence_e09b.json`.

stage A is 1.44 below NLLB-3.3B on 43 languages (36.12 against 37.56, [−2.34, −0.56]; 13 languages at least equal) and 3.67 below on all 50 languages both have; held-out AfriCOMET 0.6414 against 0.6709 (−0.0294 [−0.0366, −0.0222], 40 languages, ahead in 5). Our own NLLB run on 100 sentences is 1.23 [−1.60, −0.86] below the published full-devtest value, which bounds the sample and reference effects. AfriCOMET is the QE reward's family, so that row is not independent evidence for the RL model.

AfriqueLLM family, indicative only. On the 21 languages with per-language values in the 5-shot AfroBench-Lite harness (afr amh ewe hau ibo kin lin lug nya orm sna som sot swh tir tsn twi wol xho yor zul): SFT 32.37, stage A 39.13, B1 best 39.32, mixed best 39.40, weak final 39.26, against AfriqueQwen3.5-4B-50Langs 41.71 (mixed best −2.31; ahead in 4 of 21 languages), AfriqueGemma-12B 38.58 (+0.83; 6 of 21), AfriqueLlama-8B 36.71, AfriqueQwen-14B 36.67, AfriqueQwen-8B 35.06, AfriqueGemma-4B 31.12, Gemma3-12B 18.82, Llama3.1-8B 16.45. Those are base models in 5-shot prompting with a harness whose chrF++ implementation and FLORES split are not verified. Bidirectional aggregates (TranslatePsy, AfriNLLB) and pooled AfroBench rows are not comparable and are not compared.

Table 20: RL gain (mixed best minus SFT, FLORES+ devtest chrF++, 54 languages) by Glottolog family, script and regression family; bootstrap over languages.

grouping	group	n	mean SFT chrF++	mean gain [bootstrap CI over languages]
Glottolog top family	Atlantic-Congo	33	26.20	+9.14 [+7.58, +10.79]
	Afro-Asiatic	11	13.45	+6.89 [+4.29, +9.89]
	Nilotic	3	12.87	+11.58 [+10.30, +13.13]
	Indo-European (creoles, Afrikaans)	3	36.85	+10.23 [+1.01, +20.27]
	Mande	2	14.65	+8.42 [+7.78, +9.06]
regression family	Narrow Bantu	24	27.36	+9.91 [+7.96, +11.71]
	other Atlantic-Congo	9	23.11	+7.06 [+4.76, +9.61]
script	Latin	48	24.84	+9.32 [+8.03, +10.68]
	Ethiopic (amh tir)	2	13.55	+9.13 [+8.26, +10.00]
	Arabic (aeb ary arz)	3	4.00	+2.73 [+1.36, +4.59]
	Tifinagh (zgh)	1	2.29	+1.78

J Family, script and resource level (exploratory, fragile)

We labelled all 125 registry languages with Glottolog 5.3 (Hammarström et al., 2026) (CLDF release v5.3, commit 072ca0d of glottolog-cldf, CC BY 4.0, Zenodo 10.5281/zenodo.18840967) and asked whether the RL gain (mixed best minus SFT, FLORES+ devtest chrF++, 54 languages) depends on family, script or resource level. **This analysis is observational, has 54 languages in 7 families, several with 1 to 3 members, and its main result changes with the languages included. We report it as description, not as a finding.**

(Single-language groups, Saharan knc and Austronesian mlg, are in the report. Glottolog's Nilotic group differs from the earlier "Nilo-Saharan" group of EVID Table 4 by dik versus knc, which is why the two means agree, +11.58 and +11.62.)

- The lowest gains are the Arabic-script dialects and Tifinagh, the languages where the SFT's output was English, Arabic or loops and the RL did not repair it (Section 5.2). The eight languages that lacked a family label are now labelled: aeb, ary and arz Afro-Asiatic (Semitic, Arabic script), taq and zgh Afro-Asiatic (Berber), cjk Narrow Bantu, dik Nilotic, mfe French-lexifier creole.
- **The gain is not explained by the token budget** (Spearman rho(gain, log10 SFT tokens) = −0.16 [−0.44, +0.13], n = 54; OLS slope +0.34 chrF++ per log10 unit [−2.03, +1.79], R2 0.003), while the SFT level is (rho = +0.68 [+0.47, +0.82]; slope +11.8 [+8.5, +17.0], R2 0.45). **This is fragile:** dropping Afro-Asiatic turns the slope on the gain from +0.34 into −3.00 (SE 1.06), so "no relation" depends on which languages are in the sample. What the gain does track is the SFT's failure rate (rho with the SFT finished rate −0.47 [−0.74, −0.15], with its repeat rate +0.46 [+0.13, +0.72], n = 54 each), in line with Section 5.2: family, script and resource effects on the raw gain are largely effects on how badly the SFT failed.
- A regression of the gain on family, log tokens, non-Latin script, AfriCOMET availability and NLLB-mined share (n = 53, 11 parameters) explains little (R2 0.27, adjusted 0.09). Only non-Bantu against Narrow Bantu Atlantic-Congo is clearly non-zero (−4.45 [−8.36, −1.24], robust p = 0.020, uncorrected for the many contrasts); the NLLB-mined share is positive (+7.57 [+0.91, +14.67], p = 0.066), which reflects that NLLB-dependent languages are the low-resource tail that started low and failed by loops; it does not say whether NLLB data helped or hurt (Appendix M). The same model explains most of the SFT level (R2 0.67, adjusted 0.59). Family, script, tokens, resource tier and AfriCOMET availability overlap (VIF 2.1 to 3.4), and three of six mixed-model fits did not converge.
- Only 54 of 125 languages (58 with NTREX and MAFAND) have a gain; 51 have no reference test, so no family or script statement is possible for them. The 54 are the better-resourced languages (median 25.4 M tokens against 8.6 M for the Latin-script registry). The token table is provisional, the script of 51 languages is assigned from prior knowledge, and the analysis was run after the results were known.

K Science and document results: details

Start: the mixed-stage best. Science SFT: AdamW 3e-6, 283 steps of 65,536 tokens (6.34 M AfriScience tokens in 6 languages, two passes, plus 12.19 M replay tokens), AfriScience dev loss 1.699 → 1.322. The SFT end point (alpha 1.0) loses 2.45 NTREX and 1.20 MAFAND chrF++, so the weights were interpolated, theta(alpha) = (1 - alpha) theta_start + alpha theta_science, with a rule fixed beforehand

Table 21: Weight interpolation sweep of the science SFT, AfriScience-MT test (macro over six languages), with NTREX and MAFAND-MT chrF++ (last two columns). SSA-QE = SSA-COMET-QE (headline); AfriC. = AfriCOMET-STL.

α	SSA-QE	chrF	chrF++	BLEU	spBLEU	AfriC.	NTREX	MAFAND
0	0.6130	45.40	42.44	17.72	21.89	0.6828	37.16	40.78
0.3	0.6128	46.28	43.40	18.73	23.25	0.6865	37.13	41.06
0.5 (chosen)	0.6118	46.63	43.80	19.27	23.87	0.6879	36.90	41.05
0.7	0.6092	46.56	43.81	18.98	23.83	0.6859	36.43	40.83
1.0	0.5986	44.92	42.30	16.78	20.87	0.6708	34.71	39.58

Table 22: Science RL without the science SFT (E-14): AfriScience-MT test, sentence level, macro over 6 languages.

model	chrF	chrF++	AfriCOMET-STL (held out)	SSA-COMET-QE (reward family)
mixed best (= alpha 0; E-14's start)	45.40	42.44	0.6828	0.6130
B1 final (stage-B1 RL)	45.45	42.50	0.6846	0.6141
E-14 final (science RL, no science SFT)	**45.66**	42.69	**0.6850**	0.6153
science SFT alpha 0.5	46.63	43.80	0.6879	0.6118
science RL final (from the science SFT)	47.10	44.22	0.6932	0.6158

(best AfriScience dev chrF subject to NTREX and MAFAND not more than 0.5 below the start); alpha = 0.5 was chosen.

The science SFT gives surface gains (chrF +1.2, BLEU +1.5, spBLEU +2.0 at alpha 0.5) and no SSA-COMET-QE gain (dev 0.6242 against 0.6235). SSA-COMET-QE here is the reference-free `McGill-NLP/ssa-comet-qe`, the variant that reproduces the AfriScience-MT paper's released values (mean absolute difference 0.006 over the six strongest released runs, 0.031 over nine); there is no RL in this comparison, so it is independent of the reward. The metric choice was made after the alpha rule (Appendix R). The 385M with the science SFT is below the old 270M science-stage models on AfriScience test chrF (46.6 against 47.9 for the old RL-final, 50.0 and 50.2 for the old science-stage models) and below the published fine-tuned NLLB-1.3B (52.3).

Science RL. The run executed 160 steps (cap extended from 150 to 190 at step 120, stopped on a dev plateau at step 150 plus a 10-step wind-down; Appendix D, ledger X24). Evaluation was fixed in advance: independent evidence = plain chrF on AfriScience-MT test and held-out AfriCOMET; SSA-COMET-QE and SSA-COMET-MTL are reward-family and shown but not counted. The final checkpoint is a small real sentence-level gain over its start (science SFT, alpha 0.5): chrF +0.46 [+0.27, +0.65] on AfriScience test and held-out AfriCOMET +0.0053 [+0.0036, +0.0071]; the step-75 `best` checkpoint is smaller (chrF +0.16 [−0.04, +0.37]); SSA-COMET-QE moves by +0.004 only; NTREX and MAFAND do not regress (chrF++ +0.42, +0.10). There is no document-level gain: chunked window SSA-COMET-QE is 0.574 (start), 0.574 (best), 0.577 (final), and the earlier 270M work's 270M models stay ahead on sentence chrF (47.9-50.2 stored numbers against 47.10) and on chunked documents (0.593 against 0.577). One run, one seed. Numbers: `reports/eval_science_rl_385m_2026-11-04.md`. **Disclosure:** the long-document RL start model was changed from `best` to `final` after this evaluation, including its test numbers (rule refined on the full AfriScience dev set, where final beats best by +0.24 chrF and +0.0031 AfriCOMET; the test split was not used for the rule); logged in `reports/overnight_log_2026-11-04.md`. SSA-COMET-QE, AfriCOMET and chrF for every stage are tabulated below (E-05). The RL-only science arm (mixed best to science RL without the science SFT) is reported next (E-14).

Science RL without the science SFT (E-14). To ask whether the science SFT is needed, the executed science RL was repeated with the start model changed to the mixed best (= science-SFT alpha 0, the model the science SFT was built from): same seed 1234, same pool, same reward, Muon at 1e-6, KL 0.1 against the start model, 160 steps (150 plus a 10-step wind-down), pre-declared checkpoint = final. The rule was registered before the run: the science SFT is *not needed* if the lower 95% bounds of E-14 minus the science RL final on AfriScience-MT test are above −0.5 chrF and above −0.004 held-out AfriCOMET; it is *needed* if the upper bound of either is below its margin.

By the registered rule the science SFT is needed: the upper bounds are −1.25 chrF and −0.0065 AfriCOMET, both below their margins, chrF is lower in all six languages (−0.72 to −2.70) and the dev split agrees (−1.63 [−1.81, −1.44]). The science gain is carried by the science SFT (+1.23 chrF at alpha 0.5), not by the science RL: the RL alone on top of the mixed best adds +0.26 chrF [+0.12, +0.39]

Table 23: E-14 final minus X, 95% bootstrap (independent weights per language).

E-14 final minus	chrF	AfriCOMET-STL (held out)	SSA-COMET-QE (reward family)
science RL final (from the science SFT)	**-1.44 [−1.61, −1.25]**	**-0.0082 [−0.0101, −0.0065]**	−0.0006 [−0.0018, +0.0007]
science SFT alpha 0.5	−0.97 [−1.20, −0.75]	−0.0030 [−0.0050, −0.0009]	+0.0035 [+0.0022, +0.0047]
mixed best (its own start)	+0.26 [+0.12, +0.39]	+0.0022 [+0.0007, +0.0036]	+0.0023 [+0.0013, +0.0033]
B1 final	+0.21 [+0.07, +0.36]	+0.0004 [−0.0012, +0.0020]	+0.0012 [+0.0000, +0.0023]

Table 24: Chunked document decoding (sentence by sentence with best-of-4 repair), 168 test documents.

model (chunked)	SSA-COMET-QE, 4-sentence windows	chrF	AfriCOMET window	BLEU
385M start (alpha 0)	0.574	57.3	0.687	20.0
385M science SFT (alpha 0.5)	0.574	58.8	0.692	22.0
old 270M RL-final (earlier 270M work)	0.593	59.3	0.703	22.5

(+0.0022 AfriCOMET), a gain of the size at which seed noise matters (E-04), against +0.46 for the RL on top of the science SFT. The trainer's dev chrF++ stayed flat (43.50 → 43.43) while the executed science RL rose from 44.62 to 45.13, and no `best/` checkpoint was written (the dev macro never rose by the trainer's 0.2 margin), so only the final exists. **Reward-family caveat:** SSA-COMET-QE rates E-14 level with the science RL final (−0.0006 [−0.0018, +0.0007]) and above alpha 0.5 (+0.0035), while chrF and held-out AfriCOMET rate it below both: the metric the RL optimises moved without a matching move in the independent metrics, so it is not evidence of quality here. On the general benchmarks (93 cells) E-14 is +0.13 chrF++ [+0.10, +0.15] over the mixed best (FLORES+ +0.17, NTREX +0.12, MAFAND 0.00) and +0.34 [+0.31, +0.36] over B1 final, with no regression on NTREX and MAFAND (+0.12 / 0.00 against the mixed best; the executed science RL final is higher on MAFAND by 0.36); at document level it does not change chunked chrF (57.30 against 57.29 for its start) and whole-document decoding still fails (finished 0.69, loop 0.82, chrF 25.6). Caveats: one seed; one crash (a CUDA out-of-memory in the COMET reward worker at about step 38; the launcher's ladder halved the micro-batch from 12,288 to 6,144 tokens and resumed from the step-20 checkpoint, replaying steps 21 to 38, which can differ from the first pass because vLLM sampling is not bit-reproducible); the comparison with the science RL final mixes the effect of the science SFT with its alpha 0.5 interpolation (a science SFT at alpha 1.0 followed by RL was not run); 160 steps only.

SSA-COMET-QE for every stage (E-05). The stored AfriScience-MT sentence predictions of every stage were scored with `McGill-NLP/ssa-comet-qe`, held-out AfriCOMET-STL and plain chrF (test, macro over 6 languages). SSA-COMET-QE is 0.5686 for the SFT and 0.6118 (stage A), 0.6113 (B1 best), 0.6130 (mixed best), 0.6118 (science SFT alpha 0.5), 0.6135 / 0.6158 (science RL best / final) and 0.6158 / 0.6165 (long-document RL best / final) for the RL stages: the first RL stage adds +0.043 and every later stage at most +0.005 more. It is a reward-family metric for every RL row, so these numbers are not evidence of quality; chrF (36.00 for the SFT, 45.06 stage A, 45.40 mixed, 46.63 science SFT, 47.10 science RL final, 47.16 long-document RL) and held-out AfriCOMET-STL (0.6064, 0.6822, 0.6828, 0.6879, 0.6932, 0.6936) keep rising by small steps that SSA-COMET-QE does not register. Alpha 1.0 (the full science SFT) is the only row clearly worse (0.5986). One seed per stage; test-sentence bootstrap intervals of the differences are +/−0.001 to +/−0.002 (Section 5.5 tables). The numbers are for sentence level and are not comparable with the AfriScience-MT paper's document-level table.

168 test documents (28 papers × 6 languages), a document being the paper's plain-language summary rebuilt from sentences. Whole-document generation (greedy, 4,096 new tokens) fails for the 385M models and is worse after the science SFT: start finished 0.71, loop 0.86, chrF 22.8, tail coverage 0.27; science SFT finished 0.51, loop 0.86, chrF 24.9; the old 270M RL-final finished 0.70, loop 0.61, chrF 34.9. Sentence-by-sentence ("chunked") decoding with best-of-4 repair fixes finishing for all three (finished 1.00, sentence coverage 1.000):

The earlier 270M work's 270M RL-final is as good as or better than the 385M at document level (+0.019 window SSA-COMET-QE, +0.011 AfriCOMET window, +0.5 to +2.0 chrF; better on hau, lug, nso, zul, worse on amh). The COMET encoders read 512 tokens, so a concatenated-document COMET

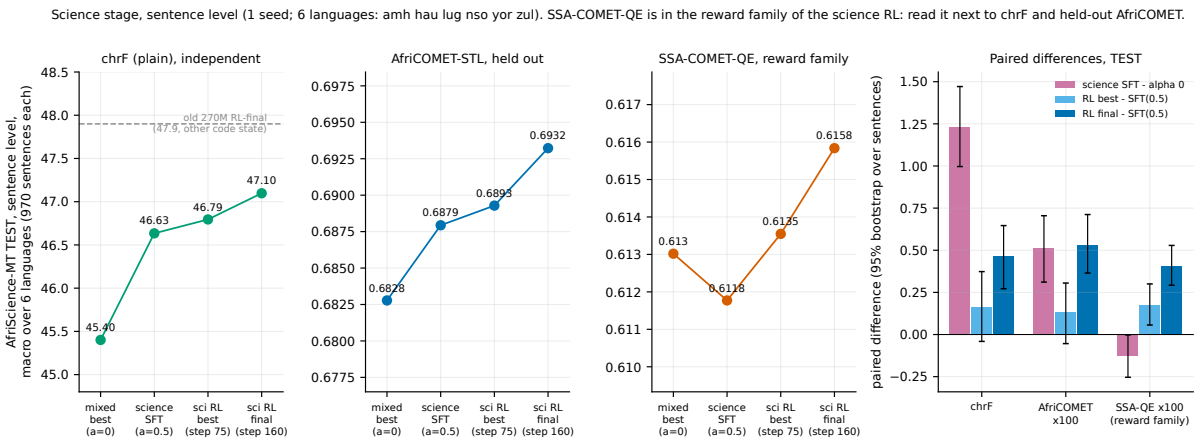


Figure 14: Science stage, AfriScience-MT TEST, sentence level (970 sentences per language, macro over amh hau lug nso yor zul; greedy; one seed): plain chrF (independent of the reward; dashed: the earlier 270M work's 270M RL-final, other code state), held-out AfriCOMET-STL (independent), and SSA-COMET-QE (reward family of the science RL) for the mixed best (alpha 0), the science SFT (alpha 0.5), and the science RL best (step 75) and final (step 160); right: paired differences with 95% bootstrap over sentences. Sources: `reports/data/eval_science_rl_2026-11-04/sci_pairs.json`, `reports/data/eval_science_sft_2026-11-04/tables.md`.

only sees the start of a document; the window values are the honest ones. Window SSA-COMET-QE is reward-family for the old model (trained against a QE reward); chrF and AfriCOMET are the independent checks. For reference, the AfriScience-MT paper's best released document-level system (GPT-5.4) has SSA-COMET 0.519 and chrF 64.5 under document handling we cannot reproduce. Caveats: one seed, 28 papers per language, no bootstrap. None of the 385M RL stages trained on prompts above 256 source tokens before the long-document RL; production use is sentence by sentence.

Long-document RL. The stage started from the science RL final (step 160), ran the planned 80 steps (32 prompts per step, about 51% long prompts of 512 to 2,048 source tokens, group 8, no guide, hold-out-reward checkpoint selection, LibreTexts-free pool) and was evaluated at its best (step 50) and final (step 80) checkpoints on the same 168 documents. In whole-document mode, best against its start: finished **0.70** $\rightarrow$ **0.82 (+0.125 [+0.046, +0.204])**, loop rate **0.85** $\rightarrow$ **0.73 (−0.113 [−0.179, −0.056])**, chrF **27.4** $\rightarrow$ **30.9 (+3.4 [+1.5, +5.3])**, tail coverage 0.32 $\rightarrow$ 0.36 (95% bootstrap over documents). The final checkpoint is similar (finished 0.85, loop 0.72, chrF 29.7, +2.2 [+0.4, +4.2], length ratio 0.81); best and final do not differ significantly (chrF −1.2 [−3.0, +0.7]). **Whole-document decoding remains unusable:** 15 to 18% of the documents do not finish, 72 to 73% show a repetition loop, and the output covers 34 to 36% of the last third of the paper (chunked: 63%). Chunked and sentence-level results do not change: window SSA-COMET-QE (reward family) 0.577 $\rightarrow$ 0.579 and 0.579 (a corpus-level number, no interval; a 0.002 difference is not an effect), chunked chrF 58.7 $\rightarrow$ 58.9 / 58.8 (best +0.13 [+0.04, +0.22]); AfriScience test sentences chrF 47.10 $\rightarrow$ 47.16 (+0.06 [−0.09, +0.20]), held-out AfriCOMET +0.0001 / +0.0004 with intervals around zero; NTREX / MAFAND chrF++ 37.33 / 41.14 $\rightarrow$ 37.43 / 41.15 (best). The earlier 270M work's 270M RL-final is still ahead in chunked mode (window SSA-COMET-QE 0.593 against 0.579) and in whole-document chrF (34.9 against 30.9). The effect is real but moderate and of the size of the other late-stage steps; the hold-out loop rate fell by only 2 points over the 80 steps, and the selection reward is the optimised quantity. One seed; 168 documents from 28 papers, so the intervals reflect paper sampling only.

K.1 Figures for the science and document results

L Contamination audit and follow-ups

6.4 Contamination. The audit scanned both sides of 25,911,095 training samples (A0 packs and the six stage-B shards) against all benchmark rows: **no benchmark sentence of 3 or more words is in the training text**; FLORES+ devtest 0 of 54,648 rows, NTREX 120 of 47,928 (0.25%), MAFAND 94 of 20,009 (0.47%), SmolSent 45 of 38,833, GlotStoryBook 15 of 3,426, all one- or two-word strings; the FLORES+ dev train half is found in 23,706 of 23,760 rows (positive control); `all`, `clean`, `no_pair` and

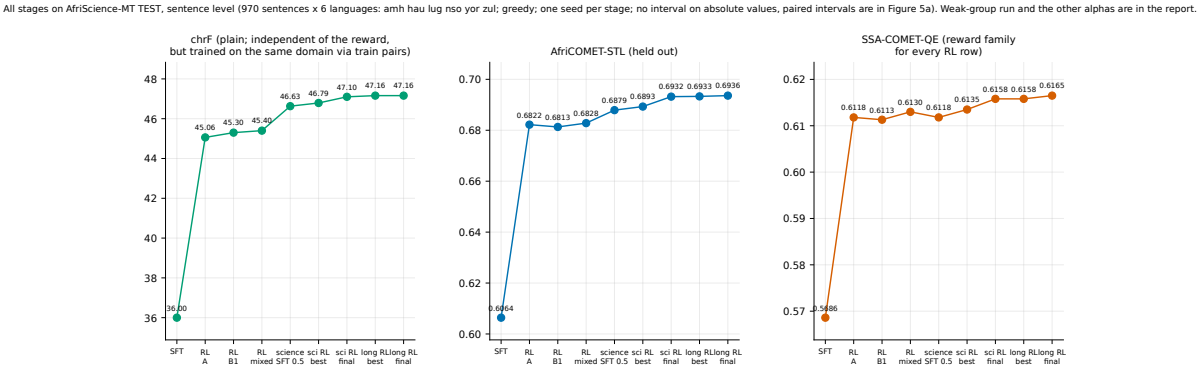


Figure 15: All stages on AfriScience-MT TEST, sentence level (970 sentences per language, macro over amh hau lug nso yor zul; greedy; one seed per stage): plain chrF, held-out AfriCOMET-STL and SSA-COMET-QE (reward family for every RL row) for the SFT, RL stages A, B1 and mixed, the science SFT (alpha 0.5), the science RL and the long-document RL. The weak-group run and the other interpolation weights are in the source table. Source: `reports/data/e05_ssa_all_stages_2026-11-04/tables.md`.

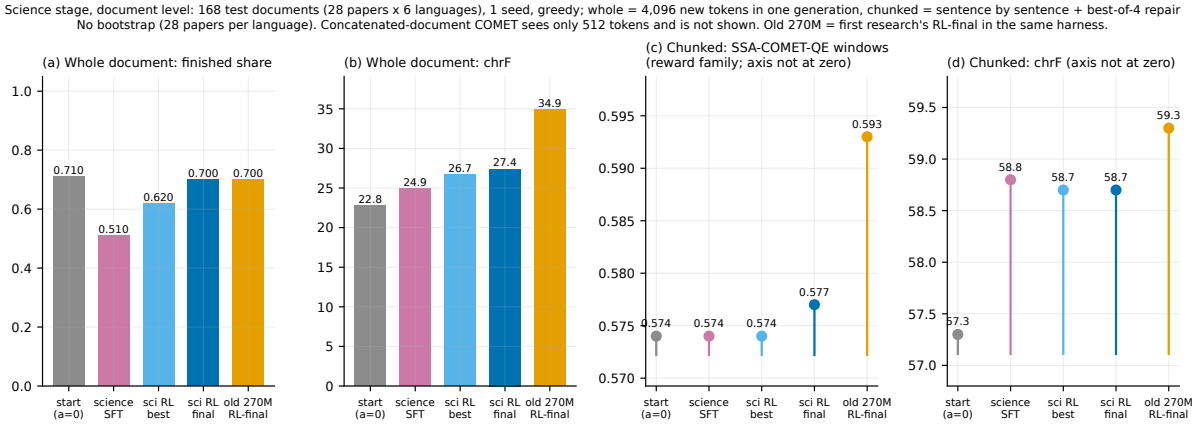


Figure 16: Science stage, document level: 168 test documents (28 papers $\times$ 6 languages), greedy, one seed, no bootstrap. Whole-document generation (4,096 new tokens): finished share and chrF. Chunked decoding (sentence by sentence plus best-of-4 repair): SSA-COMET-QE on 4-sentence windows (reward family) and chrF; axes of (c) and (d) do not start at zero. Source: `reports/data/eval_science_rl_2026-11-04/doc_tables.md`.

`clean_strict` macros agree within 0.01 chrF++.

Follow-ups (CPU, 2026-11-04).

- **Phrase level.** A lexical fuzzy scan of 56,975,939 pair rows and 4,277,612 instruction rows (reachable pair and instruction files, before gating and packing) finds no benchmark string of 8 or more tokens as a contiguous run. Strings of 4 to 7 tokens do occur inside longer training text (English side: MAFAND 134 of 17,428 strings, GlotStoryBook 30 of 898, NTREX 18 of 1,973, SALT 12 of 498; FLORES+ devtest, SmolSent, IbomMT and FLORES-MOZ 0); the audit's run rule and the intake filter test such short strings only for whole-sentence equality, so "no benchmark sentence of 3 or more words" holds for whole sentences, not for phrases. 263 (string, benchmark) cases of 8+ token English strings have a character-5-gram Jaccard ≥ 0.5 with some training text (MAFAND 184, SALT 27, NTREX 19, GlotStoryBook 15, SmolSent 6, FLORES+ devtest 4 of 1,012). These are candidates, not proof of leakage; their effect on scores was not measured.
- **Prompts of the RL.** Stage A drew 6 flagged prompts of 28,120 (0.021%); 6 of 1,345 prompts drawn for the seven languages whose prompt files had not been filtered (0.45%), all distinct, in steps 72-364, source side only (this replaces the earlier estimate of about 0.15%). The pool of the long-document RL (241,679 rows) has 0 rows flagged by the protected index.
- **AfriScience-MT train versus the SFT text.** 305 of 34,007 train rows (0.9%; hau 64, yor 116, zul 125) have their African-language sentence verbatim in the A0 pair data, all from `HuggingFaceFW/finetranslations`; 3 rows (yor 2, zul 1) as a pair. The packed shards were not scanned. The test papers are untouched (0 of 5,820 rows).
- **A0 text.** The files support that A0's pack set is the first 138,873 packs of the audited g0 (identical

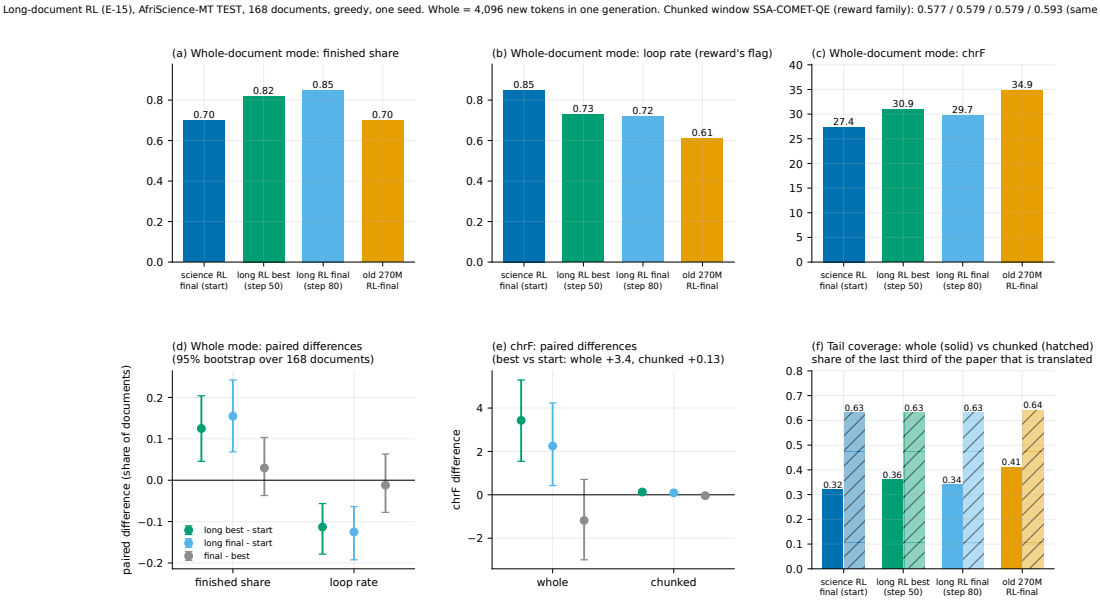


Figure 17: Long-document RL (E-15), AfriScience-MT TEST, 168 documents (28 papers × 6 languages), greedy, one seed. (a-c) Whole-document mode (4,096 new tokens in one generation): finished share, loop rate (the reward's flag) and chrF for the science RL final (the start), the long-document RL best (step 50) and final (step 80), and the earlier 270M work's 270M RL-final. (d, e) Paired differences with 95% bootstrap over documents (Poisson weights): finished share, loop rate and chrF in whole mode, chrF in chunked mode. (f) Tail coverage, the share of the last third of the paper that is translated, whole against chunked mode. Chunked window SSA-COMET-QE (reward family) is in the figure title. Sources: `reports/data/eval_long_rl_2026-11-04/{doc_tables.md,doc_pairs.json}`.

per-language pack counts for 84 languages); pack content was not hash-compared.

- **Not covered:** embedding or MT-round-trip paraphrase contamination, Gemma-3 pretraining, packed-shard coverage of the AfriScience train overlap and of the phrase scan (not run: these scans need the data box; Appendix Q item 15).

Benchmarks were evaluated many times; checkpoint choice used only the dev set (Appendix Q).

M NLLB-mined data and the regression against the earlier 270M work

All 17 languages more than 5 chrF++ below the old SFT were already below with A0 (no NLLB-mined rows); the mean gap old SFT to A0 is −10.7 and A0 to 385M SFT +2.1; Venda (no NLLB rows) regressed −9.7. Spearman of NLLB share with the regression: −0.45 [−0.73, −0.09] on NTREX+MAFAND clean (n = 28), −0.07 [−0.35, +0.20] on the pooled index (n = 57), −0.17 [−0.54, +0.23] after RL. Each factor of 10 in NLLB-mined tokens adds −0.4 chrF++ (bootstrap 95% [−11.0, +4.8]) against +9.6 for other new tokens. NLLB-mined rows are noisier than the same language's other rows (GlotLID acceptance median 0.77 against 0.99; AfriCOMET-QE below the language median in 38 of 43 languages). The decisive ablation (about 34 A100-hours) was not run [skipped by the authors; E-17]. The regression of Appendix J gives the same ambiguity: the NLLB-mined share is positively related to the RL gain and negatively to the SFT level, both through the low-resource tail.

N Negative results

- Weak-group RL (Section 5.4), science SFT on the headline metric (Section 5.5), whole-document decoding (Section 5.5), RL on the Arabic dialects and Tamazight (outside the prompt pool; English output rises, Section 5.2).
- Post-editor: an SFT post-editor applied to every sentence lowers chrF++ (FLORES+ −0.01 [−0.11, +0.10], NTREX −0.58 [−0.75, −0.41], MAFAND −0.93 [−1.14, −0.74], SmolSent −0.37 [−0.52, −0.24]); restricted to the 3.9% of drafts that are empty, unfinished or loops it gains +0.70 [+0.63, +0.78] chrF++ on FLORES+ but leaves AfriCOMET and SSA-COMET unchanged (−0.0007 to +0.0009), and beam-4 is stronger.

Table 25: Metric independence (full). IND = not in any reward; RWD = in the reward; LOOP = optimised by construction.

metric	in the RL reward?	use in this paper
`masakhane/africomet-qe-stl-1.1` (reference-free)	yes	not used as evidence
`McGill-NLP/ssa-comet-mtl` (reference-free; reference-based in the science RL)	yes	not used as evidence of RL gains
`McGill-NLP/ssa-comet-qe` (AfriScience-MT headline)	no, but the same family as the reward member	headline for SFT-only comparisons; for RL models shown only next to plain chrF and held-out AfriCOMET (all stages in Section 5.5)
TranslatePsy quality filters (AfriCOMET, SSA-COMET, MetricX)	not a reward; they filtered the training data of the E-03 control	neural metrics favour the control; chrF and chrF++ are the evidence (Section 5.3)
consensus utility (sample agreement in chrF++)	yes	chrF++ gains are partly in the optimised family
NLLB-3.3B pseudo-reference chrF++	yes, 5 languages (kea kbp nus bam kab)	chrF++ on those 5 languages is not independent; relevant to the NLLB comparison (Section 5.4)
chrF++ against a human reference, weight 0.35	yes, 7 languages (reference rows only; AfriScience TRAIN pairs)	chrF++ on those 7 languages can be partly trained; the test sets differ
language-ID, copy, repetition, length, end-of-turn guards	yes	finished and loop rates are optimised by construction
independent degeneration detector (lzma residual, repeated character 8-grams, repeated word 3-grams; thresholds from the 0.1% tail of human references)	no	measures the same idea as the loop guard with different code and constants (Section 5.2)
NLLB-218 language identification (`facebook/fasttext-language-identification`)	no	independent check of the language guard; evaluation use only (cc-by-nc-4.0)
`masakhane/africomet-stl-1.1` (reference-based)	no (held out; monitor and alarm)	the independent neural check; same family as the QE member
chrF / chrF++ / spBLEU against human references of the benchmarks	no	the main lexical evidence

O Metric independence: full table

P Per-language tables

Sources: `reports/data/rl_vs_sft_evidence_2026-11-04/tables.md` (Tables 6, 7a, 7b) and `reports/data/nllb_comparison_rl_models_2026-11-04/table2_per_language_43.csv`. Intervals are 95% paired bootstrap over sentences (Poisson weights, 1,000 resamples); NLLB columns are published values.

Table 26: RL minus SFT per language on FLORES+ devtest (54 languages, all rows; chrF++). Finished / loop are the reward's flags; "paired keeps rows where both models finished and neither loops.

iso	SFT	RL	Δ all rows [CI]	SFT finished / loop	RL finished / loop	Δ paired rows [CI]
kea	18.91	39.18	+20.27 [+18.91, +21.53]	0.83 / 0.20	1.00 / 0.00	+7.66 [+6.93, +8.42]
orm	12.06	30.18	+18.12 [+17.49, +18.76]	0.59 / 0.45	1.00 / 0.00	+2.99 [+2.51, +3.50]
nso	31.58	49.01	+17.43 [+15.75, +19.09]	0.86 / 0.15	1.00 / 0.00	+4.22 [+3.62, +4.91]
lua	17.18	33.37	+16.19 [+15.21, +17.13]	0.85 / 0.25	1.00 / 0.00	+5.45 [+4.94, +6.03]
lin	28.56	44.35	+15.78 [+14.19, +17.22]	0.90 / 0.13	1.00 / 0.00	+5.27 [+4.69, +5.84]
ssw	22.20	37.86	+15.66 [+14.46, +16.85]	0.87 / 0.17	1.00 / 0.00	+5.38 [+4.85, +5.93]
lug	16.17	31.73	+15.57 [+14.66, +16.52]	0.79 / 0.25	1.00 / 0.01	+5.32 [+4.78, +5.87]
sag	19.68	34.31	+14.62 [+13.64, +15.55]	0.87 / 0.20	1.00 / 0.00	+5.99 [+5.51, +6.47]
bem	19.31	33.88	+14.56 [+13.63, +15.61]	0.88 / 0.18	1.00 / 0.01	+7.16 [+6.60, +7.75]
tso	32.92	46.86	+13.93 [+12.43, +15.51]	0.91 / 0.11	1.00 / 0.00	+4.45 [+3.82, +5.07]
luo	21.36	34.49	+13.13 [+11.91, +14.15]	0.92 / 0.13	1.00 / 0.00	+6.44 [+5.88, +6.95]
kin	25.81	38.66	+12.86 [+11.66, +14.03]	0.89 / 0.12	1.00 / 0.00	+5.66 [+5.09, +6.21]
cjk	10.72	23.02	+12.31 [+11.70, +12.89]	0.71 / 0.31	1.00 / 0.01	+2.13 [+1.86, +2.42]
run	21.59	33.88	+12.29 [+11.21, +13.15]	0.88 / 0.15	1.00 / 0.01	+5.03 [+4.52, +5.57]
nus	13.49	24.82	+11.32 [+10.73, +11.91]	0.67 / 0.41	1.00 / 0.00	+1.97 [+1.56, +2.36]
umb	14.80	25.46	+10.66 [+9.77, +11.44]	0.89 / 0.15	1.00 / 0.01	+4.84 [+4.43, +5.26]
kmb	16.02	26.62	+10.61 [+9.82, +11.37]	0.88 / 0.23	1.00 / 0.00	+3.93 [+3.56, +4.29]
knc	3.81	14.22	+10.41 [+9.95, +10.88]	0.46 / 0.94	1.00 / 0.48	+2.49 [+0.82, +3.99]
kbp	14.51	24.88	+10.37 [+9.70, +11.00]	0.85 / 0.26	1.00 / 0.01	+5.48 [+4.98, +5.94]
kab	16.35	26.69	+10.34 [+9.50, +11.14]	0.84 / 0.20	1.00 / 0.00	+2.26 [+1.92, +2.61]
dik	3.76	14.06	+10.30 [+9.75, +10.81]	0.29 / 0.85	1.00 / 0.40	+1.80 [+1.10, +2.60]
amh	16.62	26.62	+10.00 [+9.04, +10.96]	0.87 / 0.13	1.00 / 0.00	+2.80 [+2.29, +3.29]

iso	SFT	RL	Δ all rows [CI]	SFT finished / loop	RL finished / loop	Δ paired rows [CI]
vmw	17.45	27.14	+9.69 [+8.92, +10.48]	0.84 / 0.18	1.00 / 0.00	+2.11 [+1.81, +2.44]
mfe	28.51	37.93	+9.42 [+7.98, +10.94]	0.94 / 0.07	1.00 / 0.01	+3.91 [+3.43, +4.37]
dyu	9.90	18.97	+9.06 [+8.47, +9.62]	0.83 / 0.25	1.00 / 0.00	+4.11 [+3.80, +4.44]
yor	21.32	30.33	+9.01 [+7.99, +10.00]	0.95 / 0.05	1.00 / 0.00	+5.83 [+5.09, +6.56]
tum	25.40	34.27	+8.87 [+7.89, +9.79]	0.94 / 0.08	1.00 / 0.00	+4.57 [+4.08, +5.06]
kam	17.22	25.92	+8.70 [+7.93, +9.50]	0.92 / 0.14	1.00 / 0.00	+4.42 [+4.01, +4.83]
mos	14.09	22.79	+8.70 [+8.13, +9.33]	0.90 / 0.19	1.00 / 0.01	+4.54 [+4.17, +4.89]
tir	10.48	18.74	+8.26 [+7.69, +8.86]	0.77 / 0.23	1.00 / 0.00	+2.23 [+1.83, +2.66]
taq	2.81	11.06	+8.25 [+7.82, +8.67]	0.14 / 0.91	0.99 / 0.56	+3.27 [+1.81, +4.58]
seh	21.67	29.65	+7.98 [+7.00, +8.86]	0.92 / 0.12	1.00 / 0.00	+2.01 [+1.70, +2.33]
bam	19.39	27.17	+7.78 [+6.96, +8.59]	0.89 / 0.13	1.00 / 0.00	+2.47 [+2.14, +2.80]
twi	27.54	35.07	+7.53 [+6.47, +8.61]	0.94 / 0.08	1.00 / 0.00	+2.47 [+2.02, +2.92]
tsn	37.69	45.05	+7.36 [+6.18, +8.65]	0.96 / 0.06	1.00 / 0.00	+2.09 [+1.67, +2.59]
kik	25.98	33.28	+7.30 [+6.39, +8.22]	0.93 / 0.08	1.00 / 0.00	+3.32 [+2.89, +3.81]
hau	39.88	46.46	+6.58 [+5.09, +8.34]	0.96 / 0.04	1.00 / 0.01	+0.90 [+0.50, +1.30]
mlg	39.04	44.73	+5.68 [+4.54, +6.77]	0.97 / 0.03	1.00 / 0.00	+2.03 [+1.59, +2.45]
nya	36.88	42.21	+5.33 [+4.23, +6.54]	0.97 / 0.04	1.00 / 0.00	+1.82 [+1.41, +2.24]
ewe	25.96	31.17	+5.21 [+4.46, +6.04]	0.95 / 0.06	1.00 / 0.00	+2.12 [+1.75, +2.51]
sot	37.87	42.72	+4.85 [+3.70, +6.08]	0.97 / 0.04	1.00 / 0.00	+1.15 [+0.82, +1.50]
arz	4.59	9.18	+4.59 [+4.01, +5.15]	0.49 / 0.52	0.87 / 0.19	+0.67 [+0.05, +1.25]
som	35.47	39.71	+4.24 [+3.14, +5.34]	0.97 / 0.03	1.00 / 0.00	+1.38 [+1.02, +1.75]
wol	26.87	30.58	+3.72 [+2.74, +4.63]	0.97 / 0.03	1.00 / 0.00	+1.14 [+0.82, +1.44]
sna	33.92	37.56	+3.64 [+2.47, +4.85]	0.97 / 0.04	1.00 / 0.00	+0.26 [−0.11, +0.66]
zul	45.91	48.27	+2.36 [+1.56, +3.14]	0.99 / 0.01	1.00 / 0.00	+1.26 [+0.84, +1.67]
ibo	37.82	40.14	+2.33 [+1.57, +3.08]	0.98 / 0.02	1.00 / 0.00	+0.88 [+0.53, +1.24]
ary	4.64	6.88	+2.24 [+1.76, +2.74]	0.54 / 0.48	0.97 / 0.11	−3.70 [−4.39, −3.03]
swh	56.24	58.36	+2.12 [+1.10, +3.38]	1.00 / 0.01	1.00 / 0.00	+0.88 [+0.45, +1.30]
fon	20.17	22.25	+2.07 [+1.60, +2.55]	0.98 / 0.04	1.00 / 0.00	+0.77 [+0.56, +0.97]
xho	43.56	45.43	+1.87 [+1.05, +2.80]	0.99 / 0.01	1.00 / 0.00	+0.67 [+0.28, +1.09]
zgh	2.29	4.07	+1.78 [+1.30, +2.30]	0.12 / 0.89	0.97 / 0.46	−6.69 [−9.11, −4.34]
aeb	2.75	4.12	+1.36 [+0.96, +1.78]	0.53 / 0.49	0.96 / 0.25	−2.46 [−3.19, −1.74]
afr	63.13	64.13	+1.01 [+0.53, +1.61]	1.00 / 0.00	1.00 / 0.00	+0.78 [+0.46, +1.11]

Table 27: RL minus SFT per language on NTREX (clean rows; chrF++ and held-out AfriCOMET).

iso	n	SFT chrF++	RL chrF++	Δ all rows [CI]	Δ AfriCOMET [CI]
orm	1992	9.63	27.33	+17.70 [+17.16, +18.25]	+0.168 [+0.161, +0.175]
ven	1992	21.30	38.81	+17.52 [+16.72, +18.31]	-
ssw	1992	21.58	38.80	+17.22 [+16.40, +18.02]	+0.122 [+0.115, +0.130]
bem	1992	18.68	34.79	+16.11 [+15.43, +16.84]	+0.111 [+0.105, +0.117]
nso	1992	20.14	32.73	+12.60 [+11.83, +13.35]	+0.101 [+0.095, +0.108]
tsn	1992	36.97	48.71	+11.74 [+10.68, +12.81]	+0.073 [+0.067, +0.080]
kin	1992	26.40	37.13	+10.73 [+9.95, +11.52]	+0.072 [+0.066, +0.078]
som	1992	35.84	44.23	+8.39 [+7.44, +9.25]	+0.061 [+0.055, +0.066]
yor	1992	23.77	32.10	+8.33 [+7.58, +9.10]	+0.104 [+0.097, +0.110]
tir	1992	9.10	17.00	+7.90 [+7.52, +8.25]	+0.144 [+0.136, +0.151]
amh	1992	10.58	18.05	+7.46 [+7.00, +7.93]	+0.078 [+0.071, +0.084]
ewe	1992	24.28	31.58	+7.30 [+6.64, +8.00]	+0.064 [+0.059, +0.069]
hau	1992	42.15	48.57	+6.42 [+5.24, +7.58]	+0.028 [+0.023, +0.034]
nya	1992	37.87	44.23	+6.36 [+5.46, +7.34]	+0.025 [+0.020, +0.030]
wol	1992	25.29	30.37	+5.08 [+4.34, +5.86]	+0.032 [+0.027, +0.037]
sna	1992	34.25	39.23	+4.99 [+4.17, +5.83]	+0.024 [+0.019, +0.029]
ibo	1992	39.90	44.87	+4.97 [+4.17, +5.82]	+0.028 [+0.024, +0.032]
nde	1992	28.83	32.95	+4.12 [+3.51, +4.74]	-
mlg	1992	36.91	40.85	+3.94 [+3.19, +4.74]	+0.026 [+0.021, +0.031]
swh	1992	55.18	58.73	+3.55 [+2.55, +4.54]	+0.017 [+0.012, +0.022]
zul	1992	40.92	43.50	+2.58 [+1.89, +3.29]	+0.015 [+0.010, +0.019]
xho	1992	39.85	42.36	+2.51 [+1.92, +3.16]	+0.013 [+0.009, +0.017]
afr	1992	62.42	63.92	+1.50 [+1.00, +2.15]	+0.016 [+0.012, +0.020]
shi	1992	0.24	1.16	+0.93 [+0.85, +1.02]	-

Table 28: RL minus SFT per language on MAFAND-MT (clean rows; chrF++ and held-out AfriCOMET).

iso	n	SFT chrF++	RL chrF++	Δ all rows [CI]	Δ AfriCOMET [CI]
lug	1500	26.97	39.57	+12.60 [+11.60, +13.70]	+0.065 [+0.058, +0.071]
yor	1550	23.47	33.45	+9.98 [+9.15, +10.70]	+0.127 [+0.119, +0.133]
kin	1005	36.51	46.41	+9.90 [+8.70, +11.20]	+0.064 [+0.056, +0.073]
luo	1500	22.95	32.62	+9.67 [+8.76, +10.50]	+0.072 [+0.065, +0.079]
tsn	1500	43.85	53.10	+9.24 [+7.87, +10.70]	+0.045 [+0.039, +0.052]
amh	1014	11.62	18.33	+6.71 [+5.91, +7.46]	+0.061 [+0.052, +0.070]
twi	1500	28.58	34.14	+5.56 [+4.75, +6.29]	+0.042 [+0.037, +0.047]
nya	1003	38.14	40.95	+2.81 [+1.96, +3.71]	+0.009 [+0.002, +0.015]
sna	1004	30.35	32.91	+2.56 [+1.64, +3.39]	+0.018 [+0.011, +0.025]
xho	1001	49.76	51.78	+2.02 [+1.06, +2.98]	+0.008 [+0.002, +0.014]
ibo	1458	43.55	44.89	+1.34 [+0.43, +2.32]	+0.005 [+0.000, +0.010]
swh	1822	50.68	51.79	+1.12 [+0.38, +1.89]	+0.012 [+0.007, +0.017]
zul	998	44.50	45.58	+1.08 [+0.49, +1.73]	+0.011 [+0.005, +0.017]

iso	n	SFT chrF++	RL chrF++	Δ all rows [CI]	Δ AfriCOMET [CI]
hau	1498	32.03	32.80	+0.77 [+0.21, +1.41]	+0.014 [+0.009, +0.019]
pcm	1562	53.58	53.34	−0.25 [−0.69, +0.30]	+0.007 [+0.004, +0.009]

Table 29: FLORES+ devtest chrF++ on the 43 languages shared with the published NLLB table (our values greedy; NLLB-3.3B published). Yoruba: the published value is anomalously low.

iso	language	NLLB-3.3B	SFT	RL mixed best	mixed minus NLLB-3.3B	NLLB tertile
yor	Yoruba	24.90	21.32	30.33	+5.43	tail (15)
fon	Fon	19.30	20.17	22.25	+2.95	tail (15)
dyu	Dyula	16.20	9.90	18.97	+2.77	tail (15)
wol	Wolof	28.10	26.87	30.58	+2.48	tail (15)
kam	Kamba	23.60	17.22	25.92	+2.32	tail (15)
kmb	Kimbundu	24.40	16.02	26.62	+2.22	tail (15)
umb	Umbundu	23.60	14.80	25.46	+1.86	tail (15)
afr	Afrikaans	64.70	63.13	64.13	−0.57	high (14)
ibo	Igbo	41.10	37.82	40.14	−0.96	mid (14)
nso	Sepedi	50.30	31.58	49.01	−1.29	high (14)
mos	Mossi	24.10	14.09	22.79	−1.31	tail (15)
swh	Swahili	60.00	56.24	58.36	−1.64	high (14)
lua	Luba-Kasai	35.20	17.18	33.37	−1.83	tail (15)
sag	Sango	36.50	19.68	34.31	−2.19	mid (14)
tum	Tumbuka	36.60	25.40	34.27	−2.33	mid (14)
nya	Chichewa	45.30	36.88	42.21	−3.09	high (14)
som	Somali	42.80	35.47	39.71	−3.09	mid (14)
xho	Xhosa	48.60	43.56	45.43	−3.17	high (14)
twi	Twi	38.30	27.54	35.07	−3.23	mid (14)
tso	Tsonga	50.20	32.92	46.86	−3.34	high (14)
tsn	Setswana	48.60	37.69	45.05	−3.55	high (14)
bem	Bemba	37.50	19.31	33.88	−3.62	mid (14)
lin	Lingala	48.00	28.56	44.35	−3.65	high (14)
sot	Sesotho	46.50	37.87	42.72	−3.78	high (14)
kik	Kikuyu	37.30	25.98	33.28	−4.02	mid (14)
nus	Nuer	28.90	13.49	24.82	−4.08	tail (15)
bam	Bambara	31.40	19.39	27.17	−4.23	tail (15)
luo	Luo	39.00	21.36	34.49	−4.51	mid (14)
kbp	Kabiye	29.50	14.51	24.88	−4.62	tail (15)
zul	Zulu	53.50	45.91	48.27	−5.23	high (14)
sna	Shona	42.80	33.92	37.56	−5.24	mid (14)
kab	Kabyle	32.00	16.35	26.69	−5.31	tail (15)
mlg	Malagasy	50.20	39.04	44.73	−5.47	high (14)
orm	Oromo	35.70	12.06	30.18	−5.52	tail (15)
ssw	Swati	43.50	22.20	37.86	−5.64	mid (14)
hau	Hausa	52.50	39.88	46.46	−6.04	high (14)
tir	Tigrinya	24.80	10.48	18.74	−6.06	tail (15)
lug	Luganda	38.50	16.17	31.73	−6.77	mid (14)
ewe	Ewe	38.20	25.96	31.17	−7.03	mid (14)
run	Kirundi	41.90	21.59	33.88	−8.02	mid (14)
kea	Kabuverdianu	47.90	18.91	39.18	−8.72	high (14)
kin	Kinyarwanda	48.40	25.81	38.66	−9.74	high (14)
amh	Amharic	37.90	16.62	26.62	−11.28	mid (14)

Q Limitations: full list

The numbered list of Appendix Q is a compressed version of this one.

1. **Single training seed** for every model and stage; all intervals are test-sentence bootstrap (1,000 resamples; Poisson weights shared across parallel languages for the RL-versus-SFT claims), and some are over languages only (NLLB, family analysis). Evaluation noise of the dev macro is sd 0.09 to 0.10 chrF++; training-run variation was measured only for the stage-A recipe at 100 steps on 20 languages with one pair of extra seeds (E-04: 0.06 chrF++ [−0.02, +0.14] between the two; Appendix G); seeds of later stages, of the science and long-document stages and of the SFT are not available (a second SFT seed is impractical: stage B alone is 44.8 A100-hours). The mixed step (+0.31 on FLORES+; +0.07 on the 20 languages of the seed study) and per-language differences under about 0.5 chrF++ (pcm, hau, ibo) cannot be separated from seed noise.
1. **The matched-compute control is one run.** E-03 (Section 5.3) has one seed, one learning rate (the pre-registered probe's best of three), one data mix (50% replay + 50% TranslatePsy for 19 languages, with NLLB-3.3B targets) and **one evaluated budget point (6,000 steps)**: the stage-A-matched (3,100) and mixed-best-matched (4,800) checkpoints exist but were not evaluated, at the authors' request. Its registered verdict is "inconclusive" and was not changed. The decode-time loop-guard control (E-01)

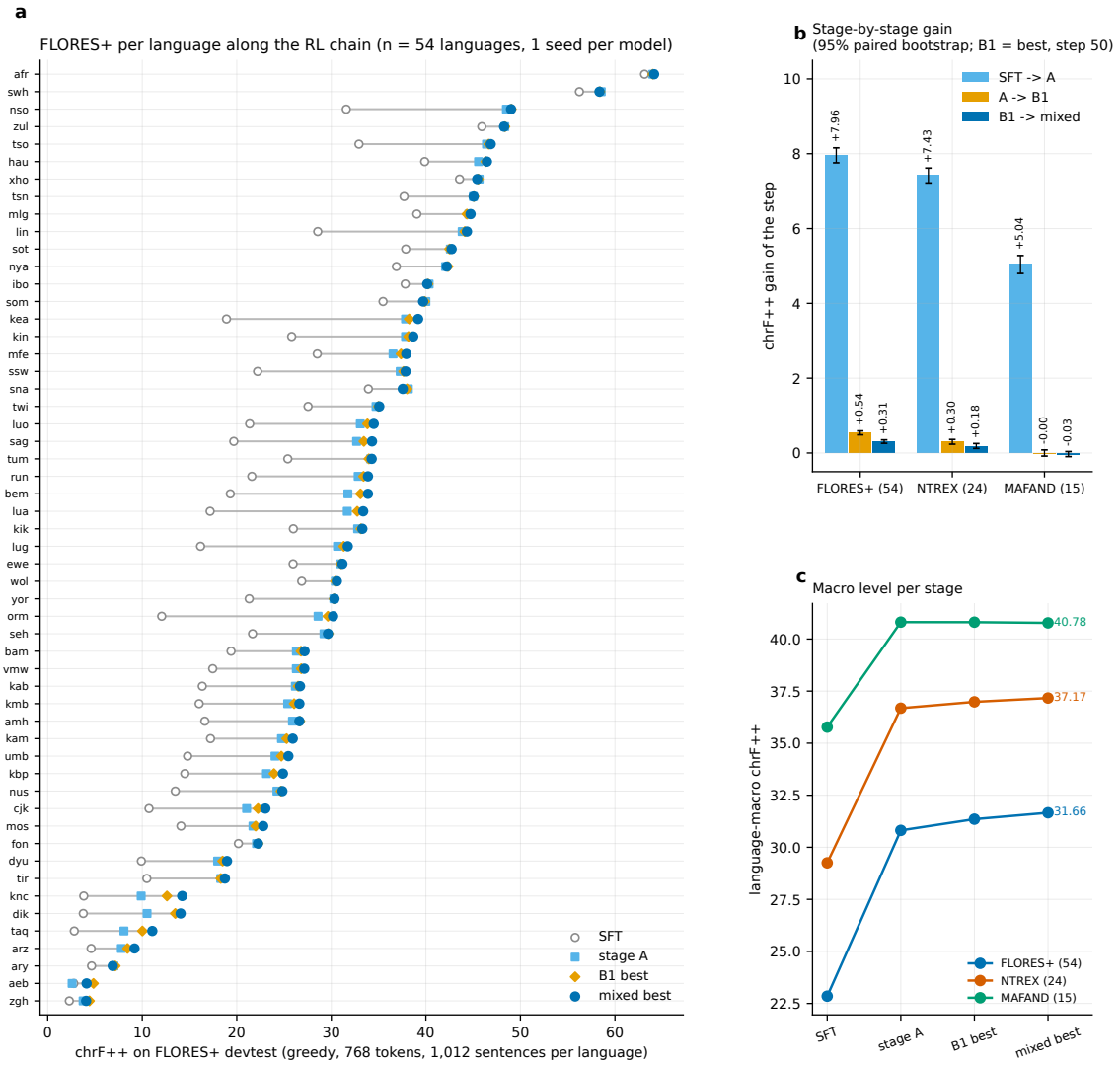


Figure 18: (a) FLORES+ devtest chrF++ per language (n = 54 languages, 1,012 sentences each; greedy, 768 tokens) for the SFT (open circle), RL stage A (square), B1 best (step 50, diamond) and the mixed best (filled circle), sorted by the mixed best; one training seed per model. (b) Gain of each RL step (language macro, 95% paired bootstrap over sentences, Poisson weights shared across parallel languages) on FLORES+ (54 languages), NTREX (24) and MAFAND (15). (c) Language-macro chrF++ after each stage. Sources: `reports/data/eval_weak_group_2026-11-03/results/results_{sft,stageA,b1best,mixedbest}.csv`, `reports/data/rl_vs_sft_evidence_2026-11-04/evidence.json`.

is reported in Section 5.2, with the caveats that its best setting was the grid edge and that a stronger penalty was explored only on dev.

2. **No human judgement of system outputs** (E-10: not run; it needs native-speaker annotators, which the authors cannot commission, and no LLM-judge substitute was validated). All evidence is automatic; the neural checks are of one family, and AfriCOMET has a length/entity bias (shown next to chrF++, never alone). One reference per sentence.
3. **Reward-family overlap.** Finished, loop and language-ID rates are optimised by construction; SSA-COMET-QE (the AfriScience-MT headline) is a checkpoint of the reward's family; chrF++ is partly optimised through the consensus utility and, for 5 + 7 languages, through the NLLB and human-reference terms. The reward's own loop flag undercounts degeneration by 1.77 on FLORES+ (Section 5.2), and the independent detector shares the repetition concept and is blind to non-repetitive failure.
4. **Ablations are limited** (E-11): three arms (no features, no reference domain, no beam guide) at 100 steps of the b64 recipe, one run per arm, sharing a 40-step prefix with seed 2, 20 languages, one pair of seeds as the noise yardstick; the KL-guard arm was not run (the guard never fired); dynamic sampling and the std floor alone, the reward components other than the reference term, and the later stages were not ablated; the mid-run changes of Appendix R mean the stages are not a clean single design.

1. **Metric choices changed after the fact for the science analysis**: the headline moved from reference-based SSA-COMET-MTL to reference-free SSA-COMET-QE after a calibration on released sentence-level runs only; the paper's own checkpoint is unverified.
2. **Engine and decoding.** The bf16 engine parity gate was not met (9.6% identical outputs vLLM bf16 against HF bf16; chrF++ 36.22, 36.80, 38.18 on 240 sentences); E-09 shows the engines agree in fp32 (98.8% and 97.1% identical outputs for the SFT and the RL model), so the gate failure is bf16 rounding. E-09b: the SFT in fp32 on the full set differs from bf16 by -0.04 chrF++ [-0.13, +0.07] over 93 cells, so the bf16 headline does not overstate the RL gain; the earlier +0.24 to +1.81 range is superseded. Not done: the RL model in fp32 and AfriCOMET scoring of the fp32 SFT run. All headline numbers are vLLM 0.32.0 bf16 greedy at 768 tokens. Beam-4 and MBR-16 were measured at 768 tokens (E-02) on the SFT and the RL model only, on 20 languages $\times$ 200 sentences, with one SFT and one RL checkpoint, chrF++ as the MBR utility (metric-matched), a primary beam definition amended after a smoke test and before any score (stop at 4 ended hypotheses), and a second beam rule that gives a different sign for the RL model; the earlier 256-token beam numbers used a loop that ran past the end token and are not comparable.
3. **Language coverage.** 51 of 125 registry languages have no reference test and are unscored; many weak languages rest on one benchmark of 48 to 1,012 sentences; the family analysis covers 54 languages. The model fails on Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri, and RL increases English output there.
4. **Comparisons.** The old-versus-385M comparison (E-13, same harness, rows clean for both generations) is still a composite of data, steps, tokenizer layout and reward, with FLORES+ reduced to 4.7% of its rows; the NLLB comparison uses FLORES-200 published numbers against FLORES+, beam against greedy and language-level intervals; AfriqueLLM numbers come from a different few-shot harness; bidirectional aggregates are not compared.
5. **Documents.** 28 papers per language (168 documents; the document intervals of the long-document RL are bootstrap over documents and reflect paper sampling only; the other document tables have no bootstrap); whole-document decoding was tested at 4,096 new tokens and remains unusable after the long-document RL (72 to 73% loops); COMET encoders read 512 tokens, so only 4-sentence-window values are document scores; the 385M RL stages trained on prompts of at most 256 source tokens before the long-document RL.
6. **Cost and scale.** One 385M model and one language registry; A100 hours as in Section 3; no scaling claim.
7. **Unresolved and unevaluated items.** B1 final (step 135) was evaluated after the fact (E-20, `reports/e20_b1_final_2026-11-04.md`): it equals B1 best (step 50) within noise (+0.01 chrF++ [-0.02, +0.04], AfriCOMET +0.0004 over 93 cells; the 85 later steps changed nothing measurable), and the mixed stage is +0.26 [+0.22, +0.30] chrF++ over B1 final on FLORES+; the NLLB-data ablation was not run [E-17]; sentence-paired comparison with NLLB on the full devtest was not run [E-06b].
8. **Not run:** human evaluation (needs native-speaker annotators); the packed-shard contamination scans, the content check of A0 against g0, the embedding and MT round-trip paraphrase scans and the source of the other 34 dev-train-half languages (need the data box); the NLLB-data ablation (skipped by the authors); beam widths other than 4, length penalties and the 54-language decoding run; a science SFT at alpha 1.0 followed by RL, a second seed of E-14 and a long-document RL from the E-14 model.

R Disclosures: full statement

Rule: data and signals that were actually used are stated, including those that are non-commercial or NLLB-derived. `DISCLOSURES.md` is the source of this section and holds the full tables.

R.1 Training data and licences

- **NLLB-derived share of the SFT.** NLLB-mined bitext (OPUS-NLLB, 43 languages, 13.46 M rows, plus a Fulah NLLB-mined set of 0.45 M rows) is **14.4% of the 3.990 B stage-B pack tokens** (13.3% of the 4.109 B executed; 10.84% of the 5.043 B tokens drawn in the whole SFT); adding NLLB-Seed (tzm)

and the aligned part of the Aya collection (NLLB-3.3B outputs; 169 M tokens in 12 languages) gives **18.7%** (746 M tokens; executed 17.2%; 14.36% of the whole SFT); adding the machine-translated Tumbuka instruction sets (system not stated, possibly NLLB) gives **22.8%** (909 M tokens; 17.86% of the whole SFT). By rows, 13.91 M of 42.31 M stage-B translation pairs (32.9%) are NLLB-mined. A0 contains none of the three. The authors allowed NLLB-mined rows with a per-language cap of 150k, ODC-BY with special attribution. NLLB models and outputs carry CC-BY-NC terms and the authors cannot use NLLB for production translation; a licence statement must accompany any release of the 385M weights. Not counted and needing a legal decision: the `*_mt560` OPUS sets (69% of the main pair output of the clean rebuild, 16.2 M of 23.4 M pairs, mostly Jehovah's Witnesses and Bible text, licence unknown, labelled NLLB-derived by an old converter docstring); `African-Languages-Lab/multi-open` (3,748,964 pairs, "license: other"); AfroLingu-MT (gated, authors' exception); other machine-translated instruction sets with unstated systems (`african_ultrachat`, `kiya_sft`, `alpaca_*`, `amharic_*`); all four families are in the instruction pack stream (`packed_instr`: every source of the instruction intake except `muri_it`, 4,190,976 samples, 1.232 B tokens; `reports/intake2_instr_packs_2026-11-01.md`); `african_ultrachat` is UltraChat machine-translated into 11 languages, `alpaca_*` (twi, swh, zul, wol, som) are translated Alpaca sets, `amharic_*` and `kiya_sft` (om, yo, sw, am rows) are machine-translated sets, and the translation system is not stated in the dataset cards or in the intake records; the packs carry no source field, so the tokens of each source drawn in stage B are not recorded.

- **Unspecified-licence sources** in the main output: KwekuAinoo/eng_twi 551,358; sentence-transformers/parallel-sentences-opus-100 210,479; UBC-NLP/AfroLingu-MT 207,796; cibfaye/wolof-english-bible 11,415; webfaq_v2; bambara_lm_qa (381k, 83% of Bambara instruction rows). Religious text dominates many low-resource languages (28 of the 81 RL languages are religious-only in the post-editor selection); the Efik NT sheet (7,936 verse pairs) has unconfirmed provenance and licence.
- **IbomMT** is CC-BY-NC-4.0: a disclosed non-commercial exception in the SFT (once) and a human-reference source of the RL reference domain (efi, ibb).
- **Quarantined and not used:** `yigagilbert/synthetic-parallel-external`, `fiifinketia/pristine-twi-english`, `muri-it`, the Aya parallel pairs.
- **Gemma base model:** `google/gemma-3-270m-it`; the Gemma terms of use apply to the derived weights; pretraining text is unknown, so exposure of benchmarks through pretraining cannot be excluded.
- **FLORES+ dev train half** (440 ids; 23,706 of 23,760 rows found in the audit) is in the SFT on purpose; the held-out half (557 ids) is protected.

R.2 RL prompts and reward-side data

- General prompts: FineWeb-Edu (ODC-By) and Wikipedia (CC BY-SA). News prompts (30% of the draw): BBC News 2017-2025 (`RealTimeData/bbc_news_alltime`, **no licence declared, BBC copyright**), `permutans/c4-bbc-news` (ODC-By label inherited from C4, BBC copyright applies), Wikinews (CC BY 2.5), Global Voices (CC BY 3.0 per the sister card), Wikivoyage (CC BY-SA 4.0; 188 prompts drawn in stage A) and `SetFit/bbc-news` (37 pool rows, none drawn, licence not recorded). The prompts are RL **input** text only; the source is never a training target.
- Seven new-language prompt files of stage A (`tll mfe guw lub crs zne lue`) had not been filtered; stage A drew 6 flagged prompts of 28,120 (0.021%), 6 of 1,345 for those languages, source side only. B1 was restarted from its step-20 checkpoint with the filtered pool.
- Reference domain (stage A from step 41, B1, mixed, science RL): human pairs from AfriScience-MT **train** (Apache-2.0; amh hau lug nso yor zul) and IbomMT train (CC-BY-NC-4.0; efi, ibb), anchors for 7 languages only. Test papers are in no pool. 0.9% of the AfriScience train rows have their African-language sentence in the A0 pair data (Appendix L).
- **NLLB-3.3B pseudo-references (CC-BY-NC) are a reward signal** for kea kbp nus bam kab (weight 0.25, max 0.35), reward only; no NLLB text is a training target in RL.
- Reward models: `masakhane/africomet-qe-stl-1.1`, `McGill-NLP/ssa-comet-mtl`; held-out monitor `masakhane/africomet-stl-1.1` (apache-2.0 verified); GlotLID language guard. Li-

cences as declared on the Hub (read 2026-11-05): `McGill-NLP/ssa-comet-mtl` and `ssa-comet-qe` apache-2.0; `masakhane/africomet-qe-stl-1.1` apache-2.0; `cis-lmu/glotlid` `license: other` with `license_name: apache-2.0-plus-notices` (card: Apache License 2.0 plus notices).

- Science stages: science RL prompts `clean_sci_v2` (PMC, arXiv CC BY/CC0, AfriScience-MT train English); **LibreTexts was excluded** from the science RL (licence conflict CC BY against CC BY-NC-SA, 10,257 prompts). **LibreTexts is not absent from the long-document RL pool**: 3,428 `common-pile/libretexts` rows (17.8% of `science_b2048`). The rows were removed before the stage started; the long-document RL did not use them. The E-14 run (science RL started from the mixed best) used the same pool and reward as the executed science RL. Science SFT: AfriScience-MT train (Apache-2.0; 28,750 windows) plus replay cut from the general SFT shards (so the NLLB, Aya and Tumbuka shares apply to the replay); 54 further catalogue candidates were scanned and none used.

R.3 TranslatePsy: used only in a research control

`qvac/TranslatePsy-AfriSLM-Synthetic-Mix` (Gritta et al.; CC-BY-NC-4.0; targets are `facebook/nllb-200-3.3B` translations of MADLAD-400 sentences, filtered by AfriCOMET, SSA-COMET and MetricX) was authorised by the authors on 2026-11-04 **for the matched-compute continued-SFT control (E-03) only**. Its English-to-X rows (19 languages: afr amh hau ibo kin lin lug mlg nya orm sna som sot swh tsn wol xho yor zul) enter that control run, scanned against the protected index (any hit on either side removed; 0.67% of the 950,000-row sample); **no TranslatePsy row is in the SFT, any RL stage, the science or long-document stages, or any released weight.** The control checkpoints are research artefacts (CC-BY-NC-4.0 and NLLB terms), are not pushed to any public repository and are not used in production. The comparison RL against control must state two things: the TranslatePsy quality filters are the reward family of the RL, so neural metrics favour the control while chrF and chrF++ do not; and TranslatePsy targets are NLLB-3.3B-like, so a good control result partly means "distilling NLLB". Status: run (preparation `reports/sft_control_prep_2026-11-04.md`; execution record and result `reports/e03_sft_control_2026-11-05.md`): 277,900 pairs per language for the 19 languages (5.28 M pairs, 96,138 packs, each seen once) after removing every row with a hit against the protected index; the control checkpoint is a research artefact under CC-BY-NC-4.0 and the NLLB terms, kept on the authors' storage, never released and never used in a released model. The comparison states that the TranslatePsy filters are the reward family of the RL, so AfriCOMET and SSA-COMET-QE favour the control, and that its targets are NLLB-3.3B outputs (Section 5.3).

R.4 Analysis decisions and data-audit facts

- The 385M SFT text contains no benchmark sentence of 3 or more words; one- and two-word strings, numerals and names occur (NTREX 0.25%, MAFAND 0.47%, SmolSent 0.12%, GlotStoryBook 0.44% of rows); phrase-level exposure is in Appendix L.
- The RL dev set is a FLORES+ dev subset (720 sentences, 18 languages). The weak-language choice used devtest-derived per-language tables (selection on a test set).
- Benchmarks were evaluated many times; checkpoint choice used only the 720-sentence dev set. The science alpha rule was written before the headline metric was fixed; the headline then changed from SSA-COMET-MTL (reference-based) to SSA-COMET-QE after the calibration. The post-hoc RL-versus-SFT decomposition ran after the results were known.
- The earlier 270M pipeline (not the clean 385M rerun) trained on text that held the English side of 95% of FLORES+ devtest; the old 270M models; their numbers appear only on their clean rows (48 sentences per language on FLORES+) and are descriptive; the earlier 270M work's FLORES+ results are not reused.
- FLORES+ and FLORES-200 references differ in places; the NLLB comparison uses published FLORES-200 numbers.

R.5 Process disclosures

- **Gates overridden by the authors.** The RL preflight gate failed (gradient norm 3.40 > 3.0, greedy loop 0.109 > 0.02, greedy finished 0.906 < 0.97) and was overridden; the stage-A gate failed (max KL 0.169

> 0.15, three KL-guard cuts, greedy loop 0.016 > 0.01) and B1 was started on the authors' instruction.
- **Mid-run changes.** Stage A was stopped at step 40 and resumed with the four features; B1 was restarted at its step-20 checkpoint (filtered prompt pool, reference domain, oversample 1.0, KL guard 0.25) and again at step 40 (MBR guide, dev early stop); the mixed stage was restarted at its step-20 checkpoint with the patch-3 monitor, and the authors loosened the hacking alarm (chrF++ 0.3 $\rightarrow$ 0.75; AfriCOMET 0.005 $\rightarrow$ 0.03). Stage A's KL guard cut the learning rate three times and it never recovered.
- **Start checkpoints.** Mixed started from B1 best (step 50), not B1 final; B1 final exists on the Hub and was evaluated after the fact (E-20): it equals B1 best within noise (+0.01 chrF++ [-0.02, +0.04] over 93 cells). The weak-group run never produced a `best/` checkpoint; its final (step 85) was evaluated. The `masakhane/africomet-stl-1.1` monitor influenced early stopping in the mixed stage and later (never a reward).
- **Incident.** The mixed stage was auto-started once by a process outside the launch agent, stopped within a minute without a step, and relaunched from B1 best. A first science-SFT pipeline pass wrote an empty alpha and launched the science RL with an invalid model path, which stopped itself at start. The GPU idled for about 2 h 10 min between stage A and B1. The E-14 run (science RL from the mixed best) had one CUDA out-of-memory in the COMET reward worker at about step 38; the launcher lowered the micro-batch from 12,288 to 6,144 tokens and resumed from the step-20 checkpoint, so steps 21 to 38 were replayed (same prompt schedule, vLLM sampling not bit-reproducible); `metrics.jsonl` holds two records for those steps and the later one is used.
- **Science RL length.** The science-RL yaml names 150 steps a hard cap, but the trainer's extension rule raised the cap to 190 steps at step 120 while dev metrics were still improving; the run then stopped on a dev plateau at step 150 and, after a 10-step wind-down, executed 160 steps (`summary.json`). The long-document RL has the extension rule off.
- **Checkpoint storage.** Stage checkpoints were written to RAM-backed `/dev/shm`; weights are in anonymised repositories, optimiser states are not.
- **Evaluation-only tool under a non-commercial licence:** the NLLB-218 language-ID model (cc-by-nc-4.0) was used for measurement only.
- **Compute.** A100 80 GB (vast.ai) for SFT stage B (44.8 h), RL (8.66 GPU-h for A + B1 + mixed), evaluation and the science stage; smaller boxes for data preparation and the post-editor. GPU-hours by component, estimated from logged durations, are in Appendix (compute table): the sized components sum to about 94 A100-hours; A0, SFT-1 and several evaluation passes are not sized; no cost figure is reported.

R.6 Intended use and risks

Outputs are machine translations of unverified quality for languages without any human evaluation, with documented failure modes: Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri are not usable (chrF++ 2 to 14 after RL), RL increases English output there, and the model should be used sentence by sentence, not on whole documents. Religious text dominates the data of many low-resource languages, which may shift register and vocabulary toward it. The model must not be presented as a replacement for NLLB-class systems: it is 3.26 chrF++ below NLLB-3.3B on 43 shared languages. The authors' use case is production translation; the licences of NLLB and TranslatePsy prevent using those systems there, which is why we state exactly where NLLB-derived text and signals entered this model.

S Reproducibility statement

- **Code and configs.** `gemma_mt/` (trainer `train_sft.py`, RL `gemma_mt/rl/`, evaluation scripts); SFT configs `reports/data/e23_hygiene_2026-11-04/sft_configs/`; RL configs `.../rl_configs/` (note: the archived stage-A `config.json` is the resumed configuration); reports and per-language tables under `reports/` and `reports/data/` as keyed in `CLAIMS_LEDGER.md`; analysis scripts for every new section in `reports/data/{nllb_comparison_rl_models,family_script_resource_analysis,degeneration_audit_independent,stage_A_feature_split,contamination_followups_385m,e23_hygiene}_2026-11-04/`.

- **Seeds and hyper-parameters.** Seed 1234 throughout; SFT: Muon, peak 5e-5 (SFT-1, shard 1) and 3e-5 (shards 2-6) with cosine floors 50% (SFT-1), 80% (shards 1-5) and 10% (shard 6), 131,072 tokens per step (Section 3); RL stages A to mixed: Muon, 2e-6, group 16, KL 0.05 (reference = each stage's start model), guard 0.15/0.25, temperature 1.0; **science RL: Muon, 1.0e-6, KL 0.1, group 16, guide 0.25, 160 steps executed (yaml cap 150, extended to 190, stopped by dev plateau); long-document RL: Muon, 1.0e-6, KL 0.1, group 8, no guide, 80 steps $\times$ 32 prompts** (Table 7); science SFT AdamW 3e-6, alpha 0.5. The science-RL yaml is in `reports/data/rl_configs_as_executed_2026-11-04/sci_rl_F_final.yaml`.
- **Hardware and software.** One A100 80 GB; vLLM 0.32.0 bf16 greedy, 768 new tokens, for every headline number. Outputs are not bit-identical across engines (9.6% identical between vLLM bf16 and HF bf16 on 240 sentences); a reader reproducing with another engine should expect small, unsystematic differences (240 sentences, HF bf16 minus vLLM bf16: +0.54 [−0.67, +1.79] chrF++ for the SFT and −0.23 [−0.73, +0.23] for the RL model; the SFT in fp32 against bf16 on the full set: −0.04 [−0.13, +0.07] over 93 cells, E-09b, which supersedes the earlier +0.24 [−0.07, +0.55] on 4,000 sentences and +1.81 on 240 sentences). sacreBLEU 2.6.0 and the chrF++/spBLEU settings are given in Section 4; the full signature strings are not recorded; exact commands per table are in the archived scripts named in the ledger.
- **Weights and branches.** RL repository `https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl`: branch `main` (mixed best, the final general RL model), `latest`, `stage-a-b64` (stage A step 200), `stage-a-final` (step 400), `stage-b1-b256` (B1 best, step 50), `stage-b1-final` (step 135), `stage-g-weak-latest`, `science-sft`, `science-sft-alpha1`, `science-rl`, `science-rl-latest`, `science-rl-final`, `long-rl`, `long-rl-latest` (long-document RL checkpoints) and tag `rl-step-50`. SFT final: `https://huggingface.co/AletheiaAING/gemma-0.4B-translate-sft` (branch `main`, step 32,853). Experiment weights: `https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl-experiments`, branches `e04-seed2-step100`, `e04-seed3-step100`, `e04-a_base-step100`, `e04-c_noref-step100`, `e04-d_noguide-step100`, `e14-science-rl-only-final`. The control checkpoint of E-03 is not published (CC-BY-NC data). No licence field or model card is declared on these repositories **[OWNER: confirm release licence]**. Run configs, logs, per-sentence outputs and scripts of E-02, E-04, E-11 and E-14 are in the research archive and the reports directory named in the ledger.
- **Data.** The decontaminated SFT and RL prompt data and the protected index (`protected-2026-10-28.3`) are in anonymised storage named in the reports and released on acceptance; the NC and unspecified-licence parts cannot be redistributed as they are (Appendix R); TranslatePsy rows are not released (Appendix R). The state, prompts, eval and reports repositories were private at the last check (Hub API 2026-11-04). [OWNER: which parts of the data are released.]
- **Long-document RL incident.** The stage took 7.7 h of wall-clock time (10:03 to 17:44 UTC), of which about 4.5 h were a crash loop: the trainer stopped (ENOSPC, root disk full) at the first step-25 evaluation and Hub-staging save and was restarted 18 times by the supervisor, each time resuming from `last/` at step 20 and repeating steps 21 to 25; the staging directories were real directories on the root disk instead of symlinks to `/dev/shm`. Fixed at 15:35 UTC; the model path of the 80 steps is otherwise the planned one (one seed). About 3.2 h were productive. The repeated step-20-to-25 segments are visible as repeated step-25 evaluations in the metrics log (Figure of the training records).
- **Not reproducible from what is stored:** stage-A steps 1-40 hyper-parameters except through the stage report; B1 final's evaluation; per-sentence NLLB hypotheses on the full devtest; the training seed spread; the SFT with another seed (stage B alone is 44.8 A100-hours).

T Prompt format and data statement

Prompt format. Every translation request is the fixed prompt `Translate the text to {language}: {english}` (two spaces after the colon) inside the Gemma 3 chat template; the model answers `<tag> {translation}<end_of_turn>` with the single-token tag of the target language; the loss covers the answer only (Section 3). **RL prompts** are English windows of consecutive

sentences from FineWeb-Edu and Wikipedia (70%) and from news (30%), filtered against the protected index; 81 languages with higher draw weights for the 29 weakest (Section 3).

Data statement. Training data are English-to-African-language pairs and instruction data in 125 languages from the sources listed in Appendix R and Appendix R; many low-resource languages are dominated by religious text; stage B includes NLLB-mined, Aya-aligned and machine-translated instruction data; evaluation data are human references of FLORES+ devtest, NTREX-128, MAFAND-MT, SmolSent, SALT, GlotStoryBook, IbomMT, FLORES-MOZ and AfriScience-MT. No personal data was collected by us; no human subjects were involved; no human evaluation was run.

U Responsible NLP checklist: items filled from the paper

- **Scientific artefacts used and licences.** Base model: Gemma-3-270m-it (Gemma terms of use apply to the derived weights). Reward and monitor models, as declared on the Hub (read 2026-11-05): `McGill-NLP/ssa-comet-mtl` and `ssa-comet-qe`, `masakhane/africomet-qe-stl-1.1` and `africomet-stl-1.1`: apache-2.0; GlotLID (`cis-lmu/glotlid`): `license: other`, `license_name: apache-2.0-plus-notices`. Training and reward-side data and their licences are listed in Ethical considerations and Appendix R: NLLB-mined and NLLB-derived data (CC-BY-NC terms), IbomMT (CC-BY-NC-4.0), AfriScience-MT train (Apache-2.0), FineWeb-Edu (ODC-By), Wikipedia (CC BY-SA), several sources with unspecified licences; TranslatePsy (CC-BY-NC-4.0, NLLB-3.3B targets) only in the research control. Evaluation uses the public benchmarks named in Section 4; the NLLB-218 language-ID model (CC-BY-NC) was used for measurement only. Release licence of the weights: the Hub repositories declare none **[OWNER: confirm release licence]**.
- **Parameters and compute.** 385,144,448 parameters; one A100 80 GB at a time. Table 30 gives A100-hours per component, *estimated from logged durations* (training seconds from the run summaries, or the logged start and end times of the job); no cost figure is reported. The sized components add up to about 94 A100-hours, a lower bound. **Not sized** (no duration in any report): the A0 stage (5,625 steps) and SFT-1 (1,500 steps); E-01 (guard grid and guarded full evaluations); E-09 and E-09b (the fp32 generation ran 17:46 to 18:38 UTC beside another job at a 0.15 GPU share); E-13 and E-20; the SSA-COMET scoring of E-05; the full evaluation passes of every stage (122,585 rows each) and the AfriScience and document evaluations; data preparation and scoring boxes; aborted runs and the compute of the earlier 270M models.
- **Number of runs and seeds.** One training run per model and stage (seed 1234); two extra seeds (20261105, 20261106) of the stage-A recipe at 100 steps; three ablation arms, one run each; the control and the E-14 run are one run each. No multiple-run statistics beyond that pair; test-sentence bootstrap intervals elsewhere.
- **Hyper-parameters and selection.** Appendix B (all SFT and RL settings, per-shard schedules, reward and features). Checkpoints were selected on a 720-sentence FLORES+ dev subset; the decode-time guard setting by a rule fixed in advance on the SFT dev grid.
- **Evaluation.** Metrics defined in Section 4; sacreBLEU 2.6.0 (from the install log of the evaluation machine; the full signature strings were not recorded), chrF++ = `CHRF(word_order=2)` with the defaults, spBLEU = BLEU with `tokenize=flores200`, NFC-normalised hypotheses and references; vLLM 0.32.0, bf16, greedy, 768 new tokens; intervals are 95% paired bootstrap, 1,000 resamples (language bootstrap, 10,000 resamples, for NLLB). Exact commands per table are in the archived scripts named in the ledger.
- **Human subjects and annotators.** None were involved; no human judgement of system outputs was collected; no personal data was collected by us.
- **Code, data and checkpoints.** See the links in Ethical considerations and the branch list in the reproducibility statement, subject to the licence limits above. NLLB-derived and non-commercial parts cannot be redistributed as they are, and TranslatePsy rows are not released **[OWNER: which parts of the data are released]**.

Table 30: A100-hours by component, estimated from logged durations (lower bound; unsized components are listed above).

component	A100-h	basis
SFT stage B (31,353 steps)	44.8	5.14 s per step
RL stage A, B1, mixed (400 + 135 + 60 steps)	8.66	15,920 + 10,479 + 4,786 s
weak-group RL (85 steps)	2.1	7,642 s
science SFT (283 steps)	0.3	17 min
science RL (160 steps)	4.8	17,153 s
long-document RL (80 steps)	7.7	wall-clock 10:03–17:44, of which about 4.5 h crash loop
E-03 control (6,000 steps and two probes)	10.5	GPU session 9.75 h, evaluation 0.55 h, SSA-COMET pass 0.2 h
E-02 beam-4 and MBR-16	2.6	logged GPU time
E-04 two seed runs and their evaluations	3.2	wall-clock 08:51–12:05
E-11 three ablation arms and evaluations	2.9	36 + 54 + 42 min of training, 3 $\times$ about 15 min of evaluation
E-14 science RL without science SFT	6.0	training 15:18–20:17, evaluation about 1 h
sized components, total	about 94	lower bound